



Sustainable Development Verified Impact Standard

IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA



Tasman Environmental Markets Asia Pacific Pte. Ltd.

Project Title	Improved Cookstove Grouped Project in Papua New Guinea
Version	1.2
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Project Location	Papua New Guinea
Project Proponent(s)	Tasman Environmental Markets Asia Pacific Pte. Ltd. Dr. Gabriel Machovsky-Capuska, Head of Science and Sustainability 15th Floor 31 Market Street, Sydney, NSW 2000, Australia Email: gabriel@tem.com.au Mobile: +61 406 419 396

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Project Lifetime	01 January 2024 to 31 December 2033; 10-year lifetime
History of SD VISTa Status	No previous attempts at SD VISTa certification made to date.
Other Certification Programs	VERRA Verified Carbon Standard (VCS) Project 4416
Expected Future Assessment Schedule	Verification anticipated in 2024

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1 SUMMARY OF SDG CONTRIBUTIONS

Table 1: Summary of Project SDG Contributions

Row number	Estimated Project Contribution by the End of Project Lifetime	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Section Reference	Claim, Asset or Label
1)	The project activity aims to replace and estimate 100,000 traditional cookstoves with fuel efficient improved cookstoves (ICS). The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices, ¹ hence reducing the economic burden of families those buy firewood under the project activity.	1.1	Goal 1: Amount of money saved by families that buy firewood	Implement activities to increase	3.2 #1	Claim
2)	Contribute to improved health and well-being brought about by reduced levels of fine particulate matter (PM _{2.5}) emissions within households below the baseline emission level of 943.3 mg/MJ delivered ² .	3.9	Goal 3: Reduction of fine particulate matter (PM _{2.5}) emissions within households to reduce air pollution within households in the project area.	Implemented activities to increase	3.2 #2	Claim

¹ ACIAR 2013_Promoting diverse fuelwood production systems in Papua New Guinea

² http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_A8WGMJX

3)	Contribute to increasing vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to stove distribution and its monitoring under VCS. An estimated 130 local staff and personnel will be trained, and training courses will be conducted annually for all the staff to make sure that the knowledge transferred is up to date and the delivery of the project remains on a high standard. At the end of each training session, staff will sit for an informal quiz to test their understanding of the training material. The measurement of improvement of learning outcomes and skill development will be based on the passing rate of the quiz. The minimum passing rate is at least 70%, and if a staff falls below the mark, they will have to re-sit the quiz within the same day.	4.3	Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Increase	3.2 #5	Claim
4)	Contribute to increasing the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs, and entrepreneurship. An estimated 130 local staff and personnel will be trained to conduct regular monitoring of the project sites for	4.4	Goal 4.2: Number of youth and adults with technical and vocational skills, by type of skill within the project area.	Increase	3.2 #5	Claim

	additional monitoring and maintenance of the cookstoves and project activity. This is alongside training in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. The local staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains on high standards.					
5)	The project will reduce women and children's drudgery through time savings in reducing time spent on cutting, collecting, and carrying firewood from trees far away from households. Monitored at 3-5 times/week in the baseline scenario as per Tasman Environmental Market's Baseline Survey Report, it is estimated to be monitored at a lower frequency (less than 5 times/week) during the lifetime of the project activity. The aim of this monitored parameter is to lower the frequency of firewood collection to 2-3 times/week.	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age, and location	Implemented activities to decrease	3.2 #3	Claim
6)	Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in the project. Monitored as the fair and unbiased employment of women within the local team during the lifetime of the project. The project envisages that 50% of its employees will be female.	5.5	Goal 5: Number of women in the project including managerial roles	Implemented activities to increase	3.2 #3	Claim

7)	Contribute to increasing access to ICS technology with Burn cookstove installations in around 100,000 households that have been using traditional three-stone fire, over the project lifetime.	7.1	Goal 7: Number of people with primary reliance on clean fuels and technology	Increase	3.2 #1	Claim
8)	With the inception of the project activity, jobs will be created firstly for the distribution of the cookstoves. Personnel will also be trained in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the cookstoves and project activity. In turn creating long term jobs for the local community, without prejudice of sex, age, occupation and persons with disabilities. An estimate of 10 full-time and 120 part-time local staff (varying length of time) will be hired during the project's lifetime, 50% of which are envisioned to be female. All the staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains on high standards.	8.3	Goal 8: Number of people formally employed, by sector and sex under the project activity.	Implement activities to increase.	3.2 #4	Claim
9)	Contribute to an increased economic income by providing fair compensation per month for the employees performing the jobs created as part of the project	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Implement activities to increase.	3.2 #4	Claim

	activity initially during deployment and later during project monitoring. Compensation will differ depending on the job scopes assigned to each formal employee employed under the project activity and will be sampled based on this distinction. Individuals employed under the project activity will likely be community members who are currently unemployed or working in the informal sector with an average income estimated as K439/month (Bonilla, et al., 2020). The project activity will compensate these employees fairly and above this rate.					
10)	Contribute to GHG emission reduction through an estimated reduction of ~2.42 tCO₂e per stove due to the replacement of baseline stoves with Burn Cookstoves over the crediting period. The above estimation is based on the project's estimated annual Emission Reductions of 242,804 tCO₂e divided by an estimated 100,000 ICSs that we planned to distribute during the project lifetime.	13.0	Tonnes of greenhouse gas emissions avoided or removed	Increase	VCS validation report	SD VISta-labeled VCU
11)	Contribute an estimated reduction in removal of woody biomass to the tune of 3.81tons of firewood per stove per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. The above value is based on the project's $B_{y,savings,i,j}$ parameter within the Emission	15.2	15.2.1 Progress towards sustainable forest management through the increase of above-ground biomass in forests	Implemented activities to increase	4.2 #1	Claim

	Reductions Calculation Sheet ³ , which relates to the quantity of woody biomass that is saved in tonnes per improved cookstove per year.					
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³ Made available to validators.

2 PROJECT DESIGN

2.1 Project Objectives, Context and Long-term Viability

2.1.1 Summary of Project Sustainable Development Objective(s)

Before the implementation of the project in Papua New Guinea, most of the local people in the island used non-renewable woody biomass for their fuel needs, for cooking with open fire or three-stone fire. The project activity will distribute fuel-efficient ICS to replace the baseline cookstoves in households. Of which the ICS disseminated through this project will replace open-fired wood-fuelled cooking methods.

The project will enable and enhance households to achieve several Sustainable Development Indicators (SDI):

The project provides ICSs with high thermal efficiency to reduce the consumption of firewood which helps achieve efficient use of an important type of nature resources, non-renewable biomass. This in turn impacts the estimated reduction in removal of woody biomass to the tune of ~3.81 tons of firewood per stove per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. (SDI15.2.1)

The project activity will replace 100,000 traditional cookstoves with fuel efficient improved cookstoves (ICS). The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices. Hence reducing the economic burden of families under the project activity. (SDI Goal 1)

The distribution of ICSs will also introduce and increase access of the local population to ICS technology with Burn cookstove installations in around 100,000 households that have been using traditional three-stone fire, over the project lifetime. (SDI Goal 7)

It will also contribute to the improved health and well-being brought about by reduced levels of fine particulate matter (PM_{2.5}) emissions within households below the baseline emission level of 943.3 mg/MJ. (SDI Goal 3)

With the inception of the project activity, jobs will be created firstly for the distribution of the cookstoves. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the cookstoves and project activity. In turn creating long term jobs for the local community, without prejudice of sex, age, occupation, and persons with disabilities. TEM envisions that 50% of the project's employees will be female. (SDI Goal 8 and SDI8.5.1)

With the creation of jobs, the project proponent will also carry out training for vocational and relevant skills of local individuals by providing non-formal education and training on issues

related to climate change, with specific skill building in operations and surveying activities related to stove distribution and its monitoring under VCS. (SDI Goal 4.1 and SDI Goal 4.2)

All in all, contributing to GHG emission reduction through an estimated reduction of ~2.42 tCO₂e per stove due to the replacement of baseline stoves with Burn Cookstoves over the crediting period. (SDG 13)

2.1.2 Description of the Project Activity

The Fuel-Efficient Cooking in Papua New Guinea project involves distribution of fuel efficient improved cookstoves (ICS) in Papua New Guinea (PNG). The ICS disseminated through this project will replace open-fired wood-fuelled cooking methods.

The project will be undertaken by Tasman Environmental Markets Asia Pacific (“**TEM**”), during which an estimate of 100,000 ICS will be distributed free-of-charge to households in project area. Tasman Environmental Markets Asia Pacific is also dedicated in creating additional value through this project, enhancing the local community’s health, well-being, and understanding and awareness of climate change and sustainable development.

The primary activity under this grouped project is the installation of ICSs.

Improved Cookstoves (G3 Burn rocket stove (also known as Kuniokoa or Ecoa wood cookstove) with adjustable pot skirt):

The primary project activity is the distribution and installation of the G3 Burn rocket stove, utilizing an air-cooled design in order to keep the inside of the combustion chamber hot and the outside cool, increasing durability, performance, and efficiency. Increased efficiency in burning woody fuel would mean less wood being used during the combustion process, encouraging local household to transition away from traditional open fire cooking to cleaner, more efficient cooking solutions.

Training:

In terms of training, Tasman Environmental Markets Asia Pacific will conduct a 2-day course⁴ for its trainers on climate change including the impacts of household pollution on health and well-being, as an introductory background to the efficient cookstove intervention. Training will generate positive impacts on community groups by enabling local community members to build alternative skills and build capacity. These interventions are expected to advance socio-economic status, generate greater community capacity for sustainable livelihoods, and increase climate resilience.

During ICS distribution, these trainers will assist stakeholders in understanding:

1. How the project activity works and its associated socio-economic benefits

⁴ Training course deck available to VVBs on request

2. How to use the ICS
3. End-User Agreement between Tasman Environmental Markets Asia Pacific and End-Users

2.1.3 Implementation Schedule

Date	Milestone(s) in the Project's Development and Implementation
31/01/2023	Consultation with Climate Change and Development Authority (CCDA) on Project Design and Local Stakeholder Consultation
31/01/2023	Consultation with PNG Forestry Authority (PNGFA) on Project Design and Local Stakeholder Consultation
13/03/2023	Local Stakeholder Consultation in Southern Highlands Province
15/03/2023	Local Stakeholder Consultation in Western Highlands Province
17/03/2023	Local Stakeholder Consultation in Jiwaka Province
20/03/2023	Local Stakeholder Consultation in Eastern Highlands Province
13/07/2023	Listing of Project under the VCS Program
13/08/2023	Undergoing validation (Public comment period closed)
01/01/2024	Estimated Project Start date: Project Activities commence with G3 Burn rocket stove (known as Kuniokoa Cookstove) distribution and installation

2.1.4 Project Proponent

Organization Name	Tasman Environmental Markets Asia Pacific Pte. Ltd.
Role in the Project	Project Developer
Contact Person	Dr. Gabriel Machovsky Capuska
Title	Head of Science and Sustainability
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2.1.5 Other Entities Involved in the Project

Organization Name	Simi Oa Consultancy Services
Role in the Project	Local consultant company for Baseline project (also known as pilot project)
Contact Person	Michael Kisombo
Title	Director
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2.1.6 Project Type

The project is categorised under type/category as below:

a) Sectoral scope: 03 - Energy demand

b) Type: II – Energy efficiency improvement projects

SD Vista Sectoral Scopes	U.N Sustainable Development Goals
1. Agriculture, forestry and other land use	Life on Land 15.2.1
2. Climate Change Adaptation	Climate Action 13.0
3. Education	Quality education Goal 4.1 Goal 4.2
4. Energy	Affordable and clean energy Goal 7
7. Health	Good health and well-being Goal 3
10. Livelihoods	No poverty Goal 1

	Decent work and economic growth Goal 8 8.5.1
14. Women's empowerment	Gender Equality 5.4.1 Goal 5

For full detail on self-defined U.N Sustainable Development goals, please refer to Section 1, Table 1: Summary of Project SDG Contributions of this document.

The project is a grouped project activity.

2.1.7 Project Location

The GPA will distribute ICS to communities within the project boundary of PNG (Latitude: 6.3150° S Longitude: 143.9555° E). The first project instance is projected to be within the Highlands Region (Figure 1), and will include the following areas: SHP, WHP, EHP and JP (see Table 2 & Figure 1). Based on the PNG census 2011, the total population size of these four Highland Provinces was 1,792,931⁵ and it is estimate that has substantially increased.

⁵ National Statistics Office, Papua New Guinea National Population & Housing Census 2011, 'Count me in' (2011)

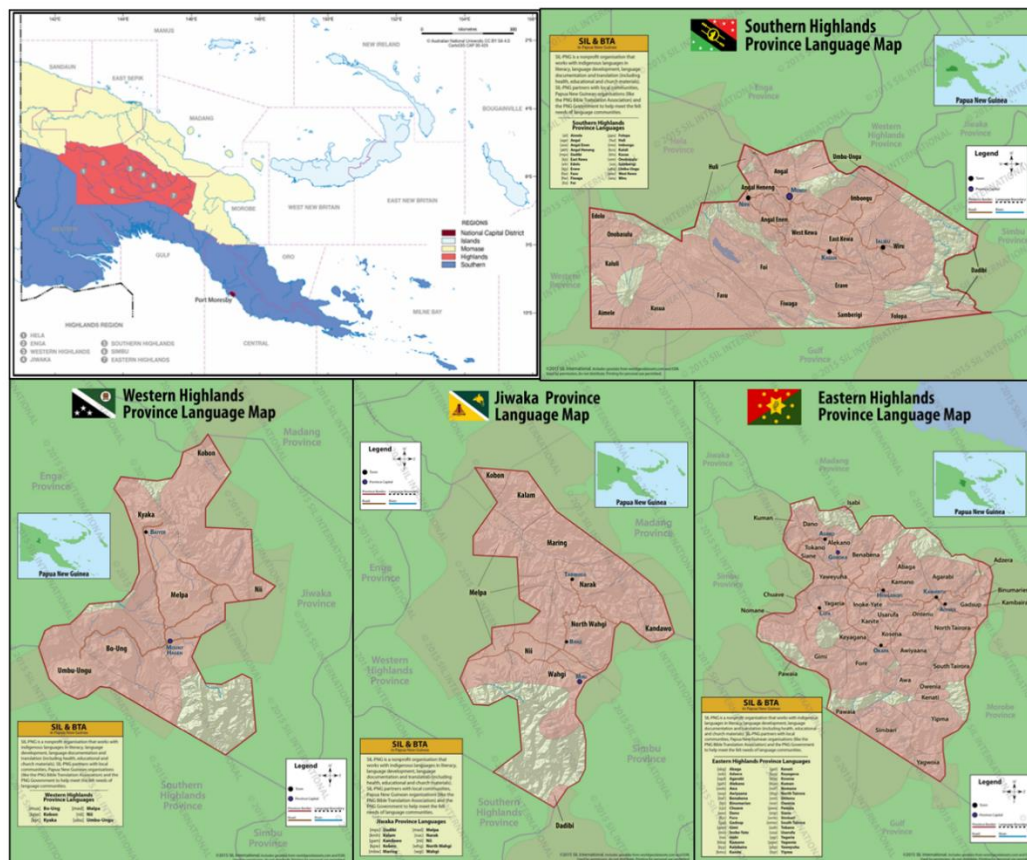


Figure 1. Map of Papua New Guinea including the Highlands Region (top left) and maps of the four different provinces included in the first project instance, Southern Highlands Province (top right), Western Highlands Province (bottom left), Jiwaka Province (bottom center) and Eastern Highlands Province (bottom right). Maps extracted from www.pnglanguages.sil.org

Table 2. The Geographic coordinates of the four provinces involved as the first project instances of the group activity in Papua New Guinea.

S/N	Location	Latitude	Longitude
1.	Southern Highlands Province	6.4179° S	143.5636° E
2.	Eastern Highlands Province	6.5862° S	145.6690° E
3.	Western Highlands Province	5.8310° S	144.1720° E
4.	Jiwaka Province	5.8564° S	144.5218° E

2.1.8 Baseline Scenario

The baseline scenario, in the absence of the project activity, showed that the target population strongly relies on wood fuel (firewood) consumption to meet their thermal energy needs mostly through traditional open fire cooking practices.

The social, economic, and natural capital conditions at the start of the project were summarized as follows:

Social Capital

Papua New Guinea (PNG) is characterised by a high degree of income inequality with 40% of people living below the extreme poverty line and 41% of children live in poverty⁶. Lack of education, healthcare and infrastructure are known as the main contributors to influence poverty in the country. In addition, 80%⁷ of the population lives in rural areas without the benefits of electrification, vaccinations, or basic improved cookstoves. It is estimated that the country's rural population increased 1.8% from 2021 to the 8,765,048 measured in 2022⁸.

The PNG population consumes over 9.40 million tonnes of firewood per year, contributing significantly to the total Greenhouse Gas (GHG) emissions for the country (ACIAR, 2013). According to the IRENA (2013)⁹, over 90.00% of the population occasionally or regularly relies on firewood for domestic and commercial cooking using open fire cooking practices. However, firewood consumption is subject to geographic location and may be bought or collected from public forests.

This was confirmed during the Baseline Study (also known as Pilot Study) conducted by Simi Oa Consulting Services on behalf of Tasman Environmental Markets Asia Pacific during October 2022 (file titled TEM 2023_Cookstoves Highlands region report_v2.pdf), where it was found that from a sampling size of 220 households, 98.00% of households commonly used firewood and only 2.00% either use electricity or gas as a source of energy.

Health and Well Being:

In accordance with the WHO's guidelines, the air quality in Papua New Guinea is considered moderately unsafe, with the annual mean concentration of PM2.5 exceeding the recommended

⁶ <https://data.unicef.org/how-many/how-many-children-live-in-poverty/>

⁷ https://www.mypsup.org/countries/Papua_New_Guinea#:~:text=Some%2080%20per%20cent%20of,country%20has%20several%20urban%20centres.

⁸ <https://www.macrotrends.net/global-metrics/countries/PNG/papua-new-guinea/rural-population#:~:text=Papua%20New%20Guinea%20rural%20population%20for%202022%20was%208%2C765%2C048%2C%20a,a%202.06%25%20increase%20from%202019.>

⁹ IRENA (International Renewable Energy Agency), 2013. Renewable energy opportunities and challenges in the Pacific Islands Region

maximum of 10 $\mu\text{g}/\text{m}^3$ ^{10,11}. Vehicle emissions, the mining industry, the oil and gas industry, and waste burning are suggested as the main drivers for this pattern. This is also magnified by indoor air pollution and smoke generated from the burning of solid fuels by traditional open fire cooking and heating that is linked to various illnesses and deaths¹².

The Baseline Study showed that open fire cooking (i.e. inefficient three-stone fire burning non-renewable woody biomass) produced thick grey or light blue smoke that led to headaches, cough, asthma, and other related diseases.

Social Impact of Firewood Collection:

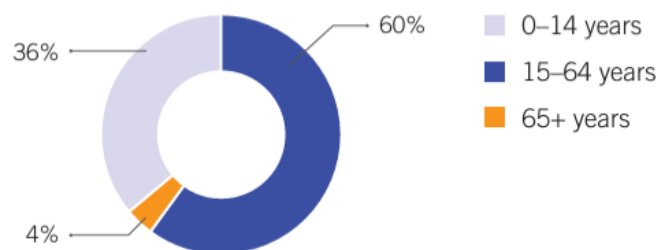
People living in rural areas across PNG were reported to collect firewood taking an average of 4 trips per week that lasted 3 hours each (ACIAR, 2013¹³). The Baseline Study showed that most of the surveyed population conduct over 3 trips per week for firewood collection.

The use of the ICS in the Highlands region potentially decreased the firewood consumption of up to 1.54 tonnes/household/meal, that in turn reduced the amount of time that families spent collecting firewood. This brings direct economic benefit for those families through purchase of less firewood, enables families to have more meaningful time for other activities, and contributes to the welfare of the local communities while conserving trees and forests that are vital to tackle the pressing issue of climate change.

Economic Capital

The population in PNG's is largely rural and growing with around 60% of the population being of working age (15-64 years) as shown in Figure 2 below:

Population age categories



Note: All data for 2016, except fertility and life expectancy, which are 2015.

Source: ILO compilation using World Bank: World development indicators, last updated 20 July 2017, <http://databank.worldbank.org> (accessed 30 July 2017).

¹⁰ World Health Organization (WHO), Health and environment scorecard: Papua New Guinea, Air Pollution

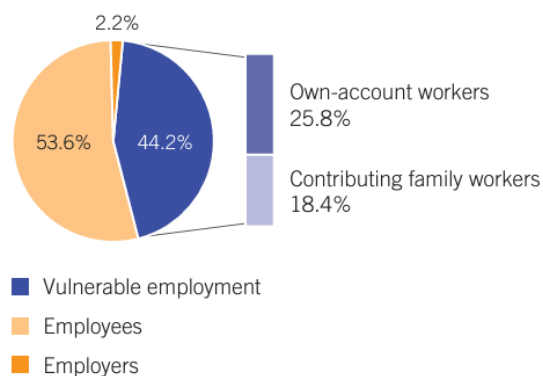
¹¹ [https://www.who.int/en/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/en/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)

¹² World Health Organization (WHO), Indoor smoke from solid fuels

¹³ ACIAR 2013_Promoting diverse fuelwood production systems in Papua New Guinea

Figure 2. Population age categories, depicting age groups that are too young to work (0-14 years), within typical working age (15-64 years), and retired (65+ years).

According to Employment and Environmental Sustainability Fact Sheet (2017), the PNG labour force participation rate was 70.5%, and the employment-to-population ratio was 68.7%, both close to gender parity. According to this study, the vulnerable employment in the country accounts for 44.2% of the labour force, with many of those workers having own-account status. The split between Vulnerable employment, Employees, and Employers is shown in Figure 3 below:



Note: Vulnerable employment includes own-account workers and contributing family workers.

Source: ILO compilation using ILOSTAT, <http://www.ilo.org/ilostat> (accessed 17 July 2017).

Figure 3. Presentation of Vulnerable Employment in Papua New Guinea by status in 2017.

Own-account and contributing family workers are more likely to experience low job and income security than employees and employers, as well as lower coverage by social protection systems and employment regulation.

Informal sector employment is particularly widespread among the youth. A gender gap in access to formal sector employment means that men are more likely to find formal jobs than women. Wages in the informal economy are considerably lower than in the formal economy¹⁴. This pattern is particularly relevant to the rural areas of the Highlands region that is characterised by low and informal income, mostly derived from the sales of agricultural produce (e.g., fresh food, coffee and firewood) in street markets (Figure 4), with 40% of the population living near the poverty line (Hanson, et al., 2001¹⁵), with an average income estimated as K439/month (Bonilla, et al., 2020¹⁶).

¹⁴ Improved Technical and Vocational Education and Training for Employment Project: Report and Recommendation of the President, <https://www.adb.org/projects/documents/png-53083-001-rrp>

¹⁵ Hanson, L., Allen, B., Bourke, R. (2001). Papua New Guinea Rural Development Handbook. Canberra, Australian National University

¹⁶ Bonilla, F., Voorend, K., Anker, R., Anker, M., & Prates, I. (2020). Anker Living Income Reference Value: Rural Papua New Guinea (2020) (No.20-03-20). Universidad Privada Boliviana.



Figure 4. Informal Street food market led by local women in the Southern Highlands Province. Photo credit: Dr. Gabriel Machovsky Capuska.

Natural Capital

Papua New Guinea is the 3rd largest island country in the world with a total area of 462,840 km², (452,860 km² land and 9,980 km² water), and a coastline of 5,152 km. Forests cover about 65% of the land area of the island (288,000 km²), with particularly high biodiversity.

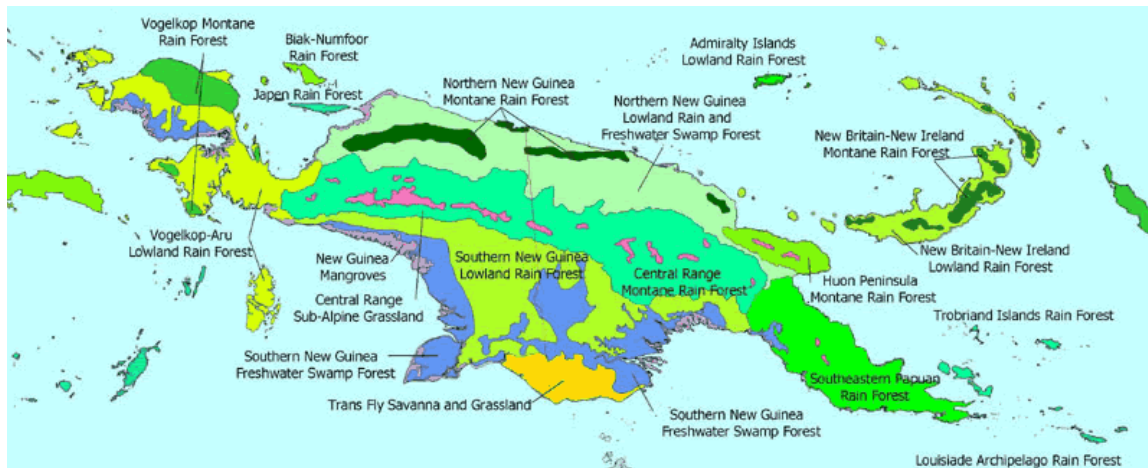


Figure 5. Forest and Eco-Regions Map of Papua New Guinea

Source: WWF: Papua New Guinea, Area of Forests

However, these forests are subject to high risk of deforestation due to commercial demand for tropical timber; forest clearance (especially in coastal areas) for plantations; and impacts from mining projects.

According to the Global Forest Watch for Papua New Guinea, from 2002 to 2023, Papua New Guinea lost 943,000 ha (2.9%) of humid primary forest, making up 51% of its total tree cover loss in the same period.

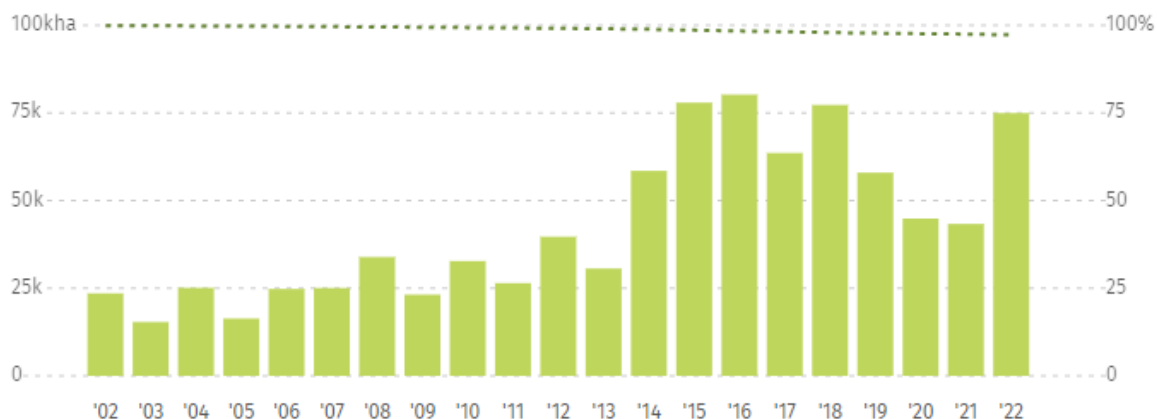


Figure 6. Primary Forest Loss in Papua New Guinea from 2002 to 2023

Source: GlobalForestWatch.org, Papua New Guinea

This was equivalent to the loss of 1.79 Mha of tree cover, equivalent to a 4.2% decrease since 2000, and 1.34 Gt of CO₂e emissions.

2.1.9 Causal Chain(s)

Please refer to Appendix I for the Causal Chain image.

Sustainable Development Outputs:

Impact No.	Description	Type of Impact		
		Outcome	Impact	Intention
1	Provide access to ICS (People and Prosperity)	The project activity aims to replace and estimate 100,000 traditional cookstoves with fuel efficient improved cookstoves.	Positive	Intended
2	Improved Health Status (People and Prosperity)	Using the ICS, gradually reduce the local people's exposure to high PM2.5 and high CO due to higher efficiency of combustion	Positive	Intended

		within their household over the lifetime of the project activity.		
3	Less time spent on unpaid domestic and care work (People and Prosperity)	Monitored at 3-5 times/week in the baseline scenario as per Tasman Environmental Market's Baseline Survey Report, it is estimated to be at a lower frequency during the lifetime of the project activity.	Positive	Intended
4	Increased Income (People and Prosperity)	Local people will be hired by the Project through a fair and unbiased process, during which we estimate that 130 jobs (10 full time and 120 part-time) will be created over the project lifespan. Of the total number of employees employed by the project, TEM envisions 50% of staff to be female.	Positive	Intended
5	Trainings imparted on climate change, project implementation and monitoring procedures (People and Prosperity)	With the inception of the project activity, vocational and technical skill development related to community engagement, survey conduction, technical trainings like conducting Water Boiling Tests (WBT) will be provided to many stakeholder groups which is envisaged to empower their lives by not only providing long-term employability but also through increased awareness levels regarding issues related to climate change, and social equity. The proposed vocational and	Positive	Intended

		technical training will also include local university students under the MoU signed with Western Pacific University (see file titled: MoU_TEM and WPU_PNG Cookstoves project_signed)		
6	Improved Economic Outcomes (People and Prosperity)	With the commencement of the project activity, the use of the ICS increases fuel efficiency, and an increase in monetary savings is expected. This change is the result of the reduction in the cost of firewood purchases over the lifespan of the project activity due to the increase in fuel efficiency with the project stoves.	Positive	Intended
7	Reduction in sales of Firewood for Firewood Vendors (People and Prosperity)	With the commencement of the project activity, it is expected that the volume of firewood sold by vendors will decrease due to a reduction in the consumption of firewood for heating and cooking during the lifespan of the project activity.	Negative	Unintended
8	Reduced demand for large diameter non-renewable firewood from live trees due to only needing finger-sized woody biomass, i.e., twigs and crop residues. (Planet)	By slowing the consumption of woody biomass for cooking purposes and by eliminating the need for large diameter woody biomass due to the technology in the ICSs, the resulting drop in demand slows deforestation, thus increasing the above-ground biomass in the adjoining forests near the project area.	Positive	Unintended

9	Avoided emission of GHGs made possible by using ICSs. (Planet)	An annual average of 242,804 tCO ₂ is estimated to be avoided due to the inception of the project activity.	Positive	Intended
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2.1.10 Threats to the Project

Natural-induced threats

Threat:	Continued deforestation and degradation make firewood an untenable source of biomass.
Solution:	As much as the efficiency of the cookstoves coupled with the imminent scarcity of the wood fuel would support the use of G3 Burn rocket stove (also known as Kuniokoa Cookstoves) as proposed by Tasman Environmental Markets Asia Pacific for this project, there is the potential that as the population continues to expand and there may be a point when firewood becomes an untenable source. In response to this, Tasman Environmental Markets in country team will train end users to maximise the use of small twigs and branches, crop residues near the local communities they live.

Human-induced threats

Threat:	Stove users' lack of upkeep of stove resulting in either malfunctioning of the stove as designed or lack of stove use.
Solution:	Tasman Environmental Markets Asia Pacific will conduct routine visits to end-user households at least twice a year in each main geographic areas that currently use the new G3 Burn rocket stove (also known as Kuniokoa), to identify potential defects in the stove's construction, maintenance, and sub-optimal use. To reduce this proposed treat, Tasman Environmental Markets Asia Pacific's 's in-country team will hire, train, and supervise users that will involve visiting the end-user's houses twice per year to monitor the use of the new ICS.
Threat:	Failure of Behaviour Change resulting in high levels of non-adoption
Solution:	Tasman Environmental Markets Asia Pacific will continuously research the impacts of behaviour change for successful adoption. Study outcomes, whether

	formal or informal, will be included in future training and community engagements to continue shaping knowledge, and enhancing positive peer-to-peer influence. Additionally, non-adopters will also be engaged to understand the potential barriers preventing them to adopt the use of the ICS. These outcomes and conclusions will be useful to improve future training events with the goal of avoiding miss-information To add to the above, Tasman Environmental Markets Asia Pacific will utilize Equation 6 of the Methodology, under which the μ_y parameter for the Bold,adjusted parameter is listed as 0.1 or 10%. Signifying the fact that 10% of end-users may potentially use both ICS as well as open fire cooking practices.
Threat:	ICS Projects and distributions are not shown to have an increased impact on household uptake.
Solution:	Co-benefits like time savings and increased health benefits aligned with the adoption of ICS work as prominent incentives for increased uptake of these stoves. In response to this, Tasman Environmental Markets Asia Pacific will conduct trainings highlighting how time saving, health benefits, previous users' positive experiences are important factors for large scale adoption.

2.1.11 Benefit Permanence

The purpose of this project is to distribute energy efficient improved cookstoves (ICS) to replace the existing practice of using traditional Cookstoves in domestic households and communities in Papua New Guinea through awareness campaigns and encouraging behavioural change to transition towards clean cooking, even beyond the life of the project.

To ensure the ICS project's long-term viability and the permanence of ICS and other program benefits, Tasman Environmental Markets Asia Pacific will take the following long-term procedures:

- 1) Periodical monitoring the end users to ensure the regular use of ICS. To guarantee the success of this process an Implementation and Spot Audit teams will be hired to conduct year-round audits. These teams will be provided with transportation and smart mobile phones that meet the physical mobility and remote connection requirements of their roles. Transportation provided will include bicycles, motorbikes, or motor vehicles, depending on their roles.
- 2) Conduct training sessions for the local communities to increase the awareness levels of these communities regarding the long-term environmental, health, and resiliency benefits aligned with the adoption of ICS. The understanding of these holistic benefits to the

stakeholders is valuable for ensuring long-term sustenance and maintenance of project activities and achieving the intended outcomes.

- 3) Tasman Environmental Markets and the Western Pacific University (WPU) signed a Memorandum of Understanding (MoU) (see file titled: MoU_TEM and WPU_PNG Cookstoves project_signed) to collaborate on the development of supporting environmental and sustainability activities directed to educate and train local students in topics related to sustainability, climate change impacts and mitigation strategies. This will provide the opportunity to educate local youth on the benefits of clean energy and clean cooking practices and to better understand the extent of the PNG improve cookstove project as a mitigation strategy for climate change.

Selected local students from WPU will gain working experience (i.e. casual, part-time or full-time employment) in different stages of project implementation in a range of transferable skills (e.g., technology IT support, administration, project management, community engagement, data collection and curation).

This partnership also contemplates the involvement of local university students and end-users designated as cookstove ambassadors in the potential development of outreach activities with primary and secondary schools in the area to foster awareness of the health, environmental and economic benefits of using improved cookstoves for cooking.

As stated in the MoU, the main goal is to involve the local communities in the proposed project activities, creating a sense of ownership amongst the people, and inspiring behavioural change from a young age through formal education.

These efforts will ensure that over the duration of the project the end users will have a positive behavioural change through the improved awareness, education and first-hand experience of the health and economic benefits over three-stone open fired cooking, which will in turn lead to the continued use of improved cookstoves even after the period of project activity.

During the lifetime of the project, TEM envisages to continue working with rural communities through periodic monitoring, training sessions and partnering with potential beneficiaries such as WPU to further secure the permanence of the behavioural changes achieved through the project activity.

Routine Spot Evidence

Spot Audit teams will adhere to the Sampling Plan carried out under the VCS Project Activity as per the Clean Development Mechanism (CDM) Guideline: Sampling and surveys for CDM project activities and programmes of activities, version 4.0, and will additionally collect and monitor the following evidence on an annual basis:

No.	SDG Target	SDG Indicator	Monitoring Frequency	Parameter Monitored	Evidence
1	1.1	Goal 1: Amount of money saved by families that buy firewood	Annually	Number of cookstoves distributed	Total number of cookstoves distributed. Total number of cookstoves distributed contributes to a reduction of firewood used, which is usually purchased at high prices.
2	3.9	Goal 3: Reduction of fine particulate matter (PM2.5) emissions within households to reduce air pollution within households in the project area.	Annually	Level of fine particulate matter (PM2.5) emissions within households' kitchen area during a cooking activity.	Sampled PM2.5 levels should be below the baseline emission level of 943.3 mg/MJ
3	4.3	Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Annually or when there are new hires	Number of participants during training sessions	Signed participation/attendance forms, onboarding documents, employment contracts.
	4.4	Goal 4.2: Number of youth and adults with technical and vocational skills, by type of skill within the project area.	Annually	Participation rate is expected to remain constant but may fluctuate slightly depending on the locals available to work.	
4	8.3	Goal 8: Number of people employed, by sector and sex, under the project activity	Annually	Employment contracts	Sampled employment contracts that can be further classified by sector and sex under the project activity.

	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Annually	Employee Payslips	Sampled Employee Payslips should show fair average earning of employees. Data to be categorized by sex, age, occupation, and persons with disabilities.
5	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age, and location.	Annually	Number of trips made to collect firewood per week	Sampled trips should below the baseline scenario of 3-5 times/week. Reduction of trips made per week indicates more time freed up to do more productive activities such as part-time work, studying, etc.
6	5.5	Goal 5: Number of women in the project including managerial roles	Annually	Number of women hired for the project activity	Employment contracts
7	7.1	Goal 7: Number of people with primary reliance on clean fuels and technology	Annually	Number of cookstoves distributed	Total number of cookstoves distributed. Total number of cookstoves distributed indicates an increase in a community's access to technology that reduces the consumption of firewood.
8	13.0	Tonnes of greenhouse gas emissions avoided or removed	Annually	Tons of CO2 removed per stove per year	Emission Reduction Calculation Sheet

9	15.2	15.2.1 Progress towards sustainable forest management through the increase of above-ground biomass in forests	Annually	Tons biomass saved per stove per year	Emission Reduction Calculation Sheet
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2.2 Stakeholder Engagement

2.2.1 Stakeholder Identification

Under this project, the stakeholders are identified as local people, communities and or representatives who are directly or indirectly affected by the activities proposed, such as end-users, stove manufacturer are identified as stakeholders.

Aligned with the United Nations Convention on the Rights of Indigenous People (2007)¹⁷ and the United Nations Development Program (UNDP) Stakeholder Engagement Guidelines (2022)¹⁸, the development of a project encompasses engagement with multiple stakeholders at different stages that spans from Project Identification to Implementation as described below.

Stage 1: Project Identification

Identification of benefits and co-benefits from project activities

The Project Identification stage is the first step of the development process and aims to identify the potential consequences of the proposed activities. These consequences are analysed, clear outcomes of the for each activity are established and linked to direct and indirect impacts to stakeholders. Those potential stakeholders affected by recognised impacts are then identified and mapped in a Causal Chain (refer to Section 2.1.9 for more details).

Stage 2: Pre-Feasibility

Determining a Project Implementation Partner

TEM appointed Simi Oa Consultancy Services, herein known as 'Simi Oa', as a Project Implementation Partner for this project activity. Simi Oa is a 100% Papua New Guinean owned company with hands-on-experience in myriad of implementing projects in the areas of education, health, climate change, and implementation of strategic plans. Simi Oa specialises in stakeholder engagement and management and has deep ties and a wide reach within the local community.

Meeting and Seeking Feedback from Government Authorities

On 08/09/2022 TEM met with the Provincial Government of the Southern Highlands Province and briefed Mr. Philip Kera, Senior staff Development Manager, and Mr. Peter Wari, Climate Change and Disaster coordinator, on the TEM's intentions to conduct the Baseline Survey within their provinces.

¹⁷ <https://www.ohchr.org/sites/default/files/Documents/Issues/IPeoples/UNDRIPManualForNHRIs.pdf>

¹⁸ https://ses-toolkit.info.undp.org/sites/g/files/zskgke446/files/SES%20Document%20Library/Uploaded%20October%202016/UNDP%20SES%20Stakeholder%20Engagement%20GN_Final_rev_July2022.pdf

The meeting with the Provincial Team emphasized TEM's engagement plan with the local communities featuring a 'bottom-up approach' that focused on the well-being and feedback of the end-users in order to maximise the impact of the project activity.

Similar information was shared with the Western Highlands Province team on 21/09/2022. Both meetings with the Provincial Government representatives led to very positive responses and overwhelming support. According to TEM's selection exercise above, the following questions were answered:

1. The identified individuals or group to benefit the most from the project activity were the households within the project boundary, also known as the 'End-Users'.
2. The only individuals or groups that would be adversely affected by the project activity were the firewood sellers within the project boundary who would likely experience lower sales upon the start of the project activity.
3. The project activity would not impact any marginalized group adversely as it aimed to distribute Improved Cookstoves to as many households as possible, free of cost.
4. The National and Provincial Authorities alongside the local communities were likely to be the main supporters of the project and no opponents of the project were identified.
5. TEM, in partnership with Simi Oa Consultancy Services would be responsible for carrying out planned activities.

Stage 3: Feasibility

Conducting Baseline Surveys

The baseline survey was held in October 2022, which involved interviewing interested household within the project boundary and conducting controlled cooking tests with them. During the baseline survey, participants were encouraged to express their points of view regarding the project and ask questions and clarifications; all this information has been integrated into the project design for improvements where possible.

The local community took great interest in the Baseline Survey and its proceedings. Feedback was continuously received during the duration of the surveying event, during which the following initial list of stakeholders were identified:

1. Local people and targeted households also known as 'End-Users'.

Group Identification: Directly Affected Group

2. Provincial Representatives and Government Officials within the project boundary.

Group Identification: Indirectly Affected Group

Following the conclusion of the Baseline Survey, TEM held a meeting with the CCDA on 31/01/2023, during which we presented the results of the baseline survey as well as the Local Stakeholder Consultation Plan. The CCDA team was also provided with the draft project

design document at that time and were asked to provide feedback on the design of the full-scale project activity.

A timeline of stakeholder engagement on both a national and provincial level prior to the local stakeholder consultation is presented below:

S/N	Meeting	Date	Supporting Documentation
1	Letter to Provincial Government to conduct the Baseline Survey Report - Southern Highlands Province (SHP)	08/09/2022	Government letter_Cookstoves Project_SHP (1) signed.pdf
2	Letter to Provincial Government to conduct the Baseline Survey Report - Western Highlands Province (WHP)	21/09/2022	Government letter_Cookstoves Project_WHP (1) signed.pdf
3	Approval of Baseline Survey to be conducted in SHP	23/09/2022	Approval Pilot study SHP.pdf
4	Approval of Baseline Survey to be conducted in WHP	27/09/2022	Approval Pilot study WHP.pdf
5	Conducting Baseline Survey in Highlands	October 2022	TEM 2023_Cookstoves Highlands region report_v2.pdf
6	Meeting with CCDA Regarding Full-Scale Project and Local Stakeholder Consultation	31/01/2023	15.VCS 4416_LOR CCDA.pdf

Conducting the Local Stakeholder Consultation

The Local Stakeholder Consultations (LSC) were publicly advertised and held in different provinces with the goal of formally introducing the project activities to the local communities and to receive their feedback.

For more information on the local stakeholder consultation, please refer to Section 2.2.3 of this document.

The LSC participation included head of households, ward councillors and educated elites. Upon conclusion of the event, it was clear that the local community could be identified and classified into different categories that included individuals or groups who were expected to have a direct or indirect impact on the project, or those who have an interest in it.

These specific categories include:

1. Local people, communities and or representatives who are directly affected (adversely affected or beneficiaries) or may have an interest in the project.
Group Identification: Directly Affected Group
2. National government officials or National Focal bodies responsible for the project in the host country, for example, Provincial Representatives and Government Officials.
Group Identification: Indirectly Affected Group
3. Local non-governmental organisations (NGOs), Women Groups working on topics relevant to the proposed activities or working with communities who are likely to be

affected by the project.

Group Identification: Indirectly Affected Group

Stage 4: Implementation

The implementation of the project activity will involve all categories of individuals mentioned above in the following ways:

- Support from Local people, communities and or representatives who are directly or indirectly affected (adversely affected or beneficiaries) or may have an interest in the project.
- Approval, support, and assistance from National and Provincial governments in the form of providing security for staff of the project activity and allowing travel within certain areas of the project activity.
- Assistance from local NGOs and Universities to perform studies where required, or to distribute Improved Cookstoves during the distribution phase of the project.

Local people will also be offered part-time and full-time jobs during the lifetime of the project activity to assist with ongoing monitoring and sampling of households at the point of project verification.

2.2.2 Stakeholders Description

As described in Section 2.2.1 above, stakeholders identified through the process can be generally grouped under three broad categories as follows:

1) The ICS end-users

The ICS end-users are local people, communities and or representatives who are directly affected or may have an interest in the project.

Before the start of the project, end-users were cooking on open fire traditional cooking practices and not aware about any type of clean cooking technology. Using ICS, less fuel (i.e., firewood) will be consumed to generate equivalent thermal energy as that in the baseline scenario. Reduction in firewood consumption will help beneficiaries in reducing expenses and saving time in cooking activities (by faster cooking).

Additionally, it will reduce the drudgery of the local people through time savings in cutting, collecting, and carrying firewood from trees far away from households as well as cooking over toxic smoke produced by open fires. The use of Improved Cookstoves would also reduce the amount of firewood required by a household for cooking and heating, directly reducing the frequency of trips made to collect firewood and allowing individuals to pursue more meaningful activities.

2) Local authorities

Local policy makers and representatives of local authorities: The project is expected to have indirect benefits for local authorities by improving the livelihoods of the people in the project

area and reducing deforestation, thereby promoting overall development and better living conditions in the region.

The list of local and provincial authorities consulted are provided in table 3 below:

Table 3: List of local and provincial authorities in support of the project activity.

Name of Authorities	Supporting Documentation ¹⁹	Date of Approval
Jiwaka Provincial Administration	Letter of No Objection (NOL)	15/02/2023
Eastern Highlands Provincial Administration	NOL	16/02/2023
Southern Highlands Provincial Administration	NOL	08/03/2023
Western Highlands Provincial Administration	NOL	13/03/2023
National Forest Authority (PNGFA)	LOA	05/05/2023
Ialibu Pangia District Development Authority	Direct Message Confirmation	14/01/2024
Imbonggu District Development Authority	Direct Message Confirmation	14/01/2024
Nipa Kutubu District Development Authority	Direct Message Confirmation	14/01/2024
Kagua Erave District Development Authority	Direct Message Confirmation	08/02/2024
Mendi Munhui District Development Authority	Direct Message Confirmation	08/02/2024
Royal PNG Constabulary	Direct Message Confirmation	15/02/2024

Direct messages as a form of confirmation were frequently used as this was the preferred method of communication amongst the District Development Authorities in Papua New Guinea.

The support from local government provides the right endorsement atmosphere that is crucial for the successful implementation of the project. It will also help the project proponent to collect feedback from end-users through the support or facilitation of local authorities.

3) NGOs

Local non-governmental organisations (NGOs), are groups working on topics relevant to the project or working with communities who are likely to be involved by the project. These organizations operate within the same geographical areas as Tasman Environmental Market Asia Pacific's projects. Their involvement would include identifying beneficiaries, managing distribution and logistics, as well as providing training, education, and capacity building. An example of NGOs engagement is represented by the Giluwe Conservation Foundation through Agnes Sumareke and Richard Kuna.

Giluwe Conservation Foundation is a historic conservation foundation launched by local communities in the precincts of Mount Giluwe in the Upper Mendi Local Level Government

¹⁹ All supporting documentation has been made available to the VVB at point of validation.

(LLG) of the Southern Highlands Province. The aim of the foundation is to protect the conservation site's forests and natural environment in the region.

Tasman Environmental Markets Asia Pacific will collaborate and work closely with the abovementioned and other local NGOs to make sure that the project will be implemented efficiently.

2.2.3 Stakeholder Consultation

The local stakeholder consultation took place in 4 Highlands Provinces, namely the Southern Highlands Province, Eastern Highlands Province, Western Highlands Province, and Jiwaka Province, on the following dates:

S/N	Province	Date
1	Southern Highlands Province	13/03/2023
2	Western Highlands Province	15/03/2023
3	Jiwaka Province	17/03/2023
4	Eastern Highlands Province	20/03/2023

Prior to the local stakeholder consultations (LSCs), Tasman Environmental Markets Asia Pacific utilized several media outlets to advertise the event to the provinces. These efforts were fulfilled using:

- Newspaper Advertisements
- Letters of Notification to Provincial Governments
- Radio Broadcasts
- Invitation Letters to Churches
- Social Media Posting
- Webinar

All advertising outlets were done in a staggered format with the aim of keeping stakeholders informed and aware of the LSCs that would be happening within their provinces. For exact dates of advertisements being published, please refer to Appendix II: LSC Outreach.

The stakeholders engaged in this project were people or groups of people that were directly or indirectly affected by the project as well as those who were interested in the project and or had the ability to influence its outcome either positively or negatively. They include:

- Representatives from the provincial administration
- Representatives from Women Groups
- Representatives from Churches
- Ward councillors

- Heads of households both male and female
- Key informants such as teachers, community health workers etc.
- Cook Stove (ICS) users (SHP & WHP only)
- Youth leaders and
- Educated elites (lecturers, nursing officers, Health Extension Officers, Police personals etc.)

Southern Highlands Province LSC

A total of 159 people attended this LSC on the 13th of March 2023 held at Kumin Catholic Mission in Mendi, SHP. Including those who commuted into Mendi from various districts, there were public servants such as nursing officers, police, teachers and church elders and pastors present during the consultation.



Figure 2. Demonstration on the use of Improved cookstoves by Ms. Magret Yangpin and Mr. Richard Manda (Field Assistants) during local stakeholder consultation at Southern Highlands Province.

The LSC was facilitated by Mr. Michael Kisombo, the project Manager with Simi Oa Consultancy Services who was engaged by Tasman Environmental Markets Asia Pacific did a power point presentation after welcoming and acknowledging all the stakeholders that were present. The PowerPoint slides were prepared in English, and explained in *tok pidgin*, the common language where most could understand well as those participated came from different local dialects.

Attendees were briefed of the purpose of the meeting, and informed of the:

- Carbon cycle and the important role it played within their forests.
- Findings of the Pilot Project conducted in SHP and WHP
- The Verra and SD VISta registration process
- The role of carbon finance in the project activity
- Indicative timeline and implementation plan of the project

Two field assistants were then invited to demonstrate and to explain how the improved cookstove is used using the sample that was exhibited during the LSC. That was further reinforced by one of the users who was also invited to give her experience on how she is currently using the improved cookstove and explained how her and her family are benefiting.

After the presentation, stakeholders were given the opportunity to ask questions and provide their reflections. For the summary of questions asked and subsequent answers provided by Tasman Environmental Markets Asia Pacific, please refer to Appendix III: Stakeholder Feedback.

After the discussions, there were about 29 feedback forms that were received through the feedback forms. These feedback forms may be made available to validators upon request.

Western Highlands Province LSC

A total of 61 people attended this LSC on the 15th of March 2023 held at Kuri Lodge in Mt Hagen, WHP. Participants who attended came from different backgrounds and electorates or districts. The LSC was also covered by Radio Eagle (local radio station) that was present all throughout the presentation.



Figure 3. Demonstration on the use of Improved cookstoves by Mr Tony Yambo (Field Assistant) during local stakeholder consultation at Western Highlands Province.

Prior to the introductions and presentation, Pastor Enoc Wan of Hagen Central – Pentecostal church opened the session with a word of prayer. Mr. Michael Kisombo then introduced the project and invited his field assistants to introduce themselves. He then proceeded with the presentation using *tok pidgin*.

He gave a brief introduction of the flow of his presentation and explained the purpose of the meeting. He then introduced the project activity before giving an overview of the carbon emissions and the forests' role in restoring carbon and its impact if most forests are destroyed. Following the background of the project, he explained the pilot project conducted in WHP and the results attained from the survey and invited end users (from the pilot project) present to explain how they are currently using the improved cookstoves to date and its impacts on them.

Two volunteers in the group gave their reflections and how they are using the cookstove. Both spoke highly of the impacts the ICS had on their families. Michael then asked Tony (one of the fields assistants) to demonstrate how the improved cookstove is used. After that presentation, Mr. Kisombo proceeded with the explanation of the Verra registration process and role of carbon finance and continued to explain the large-scale project, its indicative timeline and implementation schedule.

After the presentation, stakeholders were given the opportunity to ask questions and to give their reflections followed by a closing prayer by Pastor Enoc. There were 29 feedback forms received through the feedback forms out of the 61 participants.

Jiwaka Province LSC

More than 90 people attended this LSC on the 17th of March 2023 held at Jiwaka Mission Resort (JMR) in Banz, Jiwaka. During initial consultations to invite the provincial administrator to the LSC, the provincial administrator showed great support for the project and was able to communicate widely to his people to participate. Due to work commitments, he sent his Deputy Administrator to attend the local stakeholder consultation instead.



Figure 4. the Deputy Administrator of Jiwaka Province (Left) and in-country project manager, Michael Kisombo (right) and during local stakeholder engagement at Jiwaka Province.

Mr. Michael Kisombo led the discussions with a welcome and thanked everyone who came to Jiwaka Mission Resort for the local stakeholders meeting. He also did the presentation in *tok pidgin* – aided by the slides on power-point that was used in two other provinces. Similar to past LSCs, the topic on carbon cycle and the destruction of the ozone layer was simplified to the level of the participants. This topic was further elaborated on and supported by a few participants who had wider experience working or teaching in schools in this area of specialty.

The pilot in SHP and WHP and the coordinators of these projects, Mr. Patrick Wapiyu and Arnold Terry, were also on hand to explain the results of the pilot study and demonstrate how the improved cookstove was used. After these discussions, Mr. Kisombo continued with the presentations on the purpose of the stakeholder meeting, the verra registration process, carbon finance and the sustainable goals that this project is meeting. Towards the end, the indicative timeline and implementation schedule was explained before allowing for question-and-answer session.

There were about 29 feedback forms received through the feedback forms out of the 90 registered participants.

Eastern Highlands Province LSC

Around 29 people attended this LSC on the 20th of March 2023 held at Steakhouse Goroka in Goroka, EHP. The low turn-out rate may be attributed to the project being very new to the local stakeholders. Most stakeholders who attended learnt of the event word of mouth or through friends' network and face-book post. Despite a lesser number the discussions were fruitful. Almost all agreed that they see there is no harm and supported the project.



Figure 5. Participants reading the filling up the feedback forms during the Eastern Highlands LSC at Steak House Goroka.

Similar to the first three LSCs, the presentation was conducted by Mr. Michael Kisombo in *tok pidgin* – as the common language where most could understand well. Michael followed the same presentation he did in the other three provinces. The sample ICS that was used for this LSC was displayed and a demonstration was done on how the ICS would be efficiently used. Although it was the first presentation made to the stakeholders in the Eastern Highlands Province, everyone who attended was positive with the project and supported the project activity.

After the discussions, there were about 24 feedback forms received from the 29 participants.

At the end of the LSCs, stakeholders were informed of the project's email (pngcookstoves@tem.com.au), as well as direct contact details to the in-country project manager and the Head of Science and Sustainability in Tasman Environmental Markets Asia Pacific s. In all marketing documents, the project email and website were also published, to ensure stakeholders had multiple avenues to reach out to Tasman Environmental Markets Asia Pacific with feedback, allowing for on-going communication.

For more information, a Local Stakeholder Consultation Report is available upon request of the validator.

2.2.4 Continued Consultation and Adaptive Management

Consultation and Adaptive Management will continue throughout the project lifetime as Tasman Environmental Markets Asia Pacific is committed to best meet the needs of the beneficiaries and amplify, where possible, the co-benefits of the project activities as they relate to the SDGs.

Alongside the above, Tasman Environmental Markets Asia Pacific aims to explore additional projects within the region for the implementation of these activities. This is to multiply sustainable development and improve health and economic wellbeing within the project area.

To better adapt to the changing landscape of the project region, Tasman Environmental Markets Asia Pacific will collect information from its local team twice a year. Interactions may range from informal consultation discussions to more formal surveys with specific questions on use and adoption. Some of the visits are conducted that inquire about likes, dislikes, adoption rates, benefits that Tasman Environmental Markets Asia Pacific may not have identified, and unidentified challenges for the cook.

2.2.5 Anti-Discrimination

Tasman Environmental Markets Asia Pacific will distribute ICSs to local people who uses traditional low efficiency cooking solutions without distinction in genders, races, religions, educational backgrounds, or any other aspects. Tasman Environmental Markets Asia Pacific is dedicated to the elimination of discrimination, and as jobs are created during the implementation of this project, the company will ensure that women, persons with disabilities, and minority will have equal chance to get these jobs.

2.2.6 Worker Training

The implementation of project needs plenty of local people to participate in production, distribution or use steps. They will get relevant skills and sustainable development and global citizenship education through training by project proponent. Besides technical skills, the workers will be trained about the management structure, regulations, and worker rights.

Tasman Environmental Markets Asia Pacific will organise a two-day training course for the in-country team to learn about monitoring techniques including survey methods, data record and analysis. The monitoring plan will be carried out by qualified personnel trained for quality assurance and quality control.

2.2.7 Equal Work Opportunities

Tasman Environmental Markets Asia Pacific is dedicated to make sure that all stakeholders, no matter their genders, races, religions, educational backgrounds, or any other aspects, will be given an equal opportunity to fill all work positions. Recruitment advertising for the local villages will be given before the hiring of the full-time and part-time staff for the project. The recruitment advertisement will be provided in Pidgin English for them to understand.

2.2.8 Workers' Rights

Tasman Environmental Markets Asia Pacific will have a labour contract with each in-country personnel following PNG labour legislation. Before the workers sign it, the contracts will be explained explicitly to make sure that they could understand their rights and obligations. The contract terms will conform with, and uphold the principles and rights of, work addressed in the Core Labour Conventions of the International Labour Organization (ILO), as well as comply with national and local labour laws within Papua New Guinea.

2.2.9 Occupational Safety Assessment

To mitigate potential occupational safety hazards, Tasman Environmental Markets Asia Pacific makes the effort to contract workers who have lived in the community for several years, thus assisting in mitigating the occupational hazards through community familiarity, language fluency, and being native to the culture. This understanding of traditional values, respect, and working environment in the communities Tasman Environmental Markets Asia Pacific serves to support reduction of safety hazards.

Risks are assessed and mitigated where possible and workers, stakeholders, and others directly involved with the projects are made aware of the potential risks and opportunities to mitigate where possible, yet it should be noted that employment or use of the stove does not increase the safety risks that are already present in the landscape of the country.

2.2.10 Feedback and Grievance Redress Procedure

During the Local Stakeholder Consultations, it was discussed how stakeholders may keep in contact with Tasman Environmental Markets Asia Pacific's in country project manager, Michael Kisombo. During which, the following solutions were offered:

1. Reaching out to the in-country Team during the twice a year visits to maintain the ICSs.
2. Reaching out to Tasman Environmental Markets Asia Pacific through the project's dedicated email.
3. Reaching out to Tasman Environmental Markets Asia Pacific through the project's website.

The following website of the project activity was available to the stakeholders: <https://www.tasmanenvironmental.com.au/pngcookstoves/>

The following email addresses will be also used for inquiry or comment on the project: pngcookstoves@tem.com.au

Tasman Environmental Markets Asia Pacific in-country Project Manager: Michael Kisombo, +675 7262 3864, mkisombo@hotmail.com and currently michael.kisombo@tem.com.au.

The Grievance Redress Mechanism (GRM) is a community driven approach built on traditional Melanesian customary protocols, that are formalised to accept, assess, and resolve feedback or complaints regarding the implementation of the project.

The different channels to report grievances are summarised in Table 4, whereas the process of organising common types of grievances are captured in Table 5. The Procedure recognises and prioritises the traditional and customary means of conflict resolution in Melanesian society by encouraging dialogue through clan leaders and staff at the community level, and utilising Field Assistants who are sourced from local communities to act as the first responders while addressing grievances. Outcomes of the GRM are recorded in the GRM logbook. These measures are taken to enhance Project accountability and transparency and to support the project initiatives.

Table 4: Channels for Grievance Reporting

Grievances Reporting Channel	Grievance Reporting Mechanism
Physical walking in	Complains to TEM PNG Office and/or its staff
Social Media and online	Complains made online to TEM PNG Office, and/or in country Project manager
Phone call	Phone calls to Field Assistance, and/or Project manager
Verbal through community representatives	Complains through community representatives who then channel through the organizational structure

Table 5: Process of organising common grievances

Type of Grievance	Scope	How will be facilitated	Who will handle	Timeline
Unfair distribution of ICS	Cookstoves unfairly distributed by project staff	Gather information, investigate and refer to project lead	In country Project manager	7 -14 days
Burglary, sold or destroyed by natural disaster	Cookstoves were stolen, destroyed, or sold	Gather information, investigate, and refer to project lead	Project manager and project lead	14 – 21 days
Household missing out on the distribution	Cookstoves missed targeted households	Investigate and refer to project lead for possible re-issue	As above	5 – 7 days
Damaged cookstoves	Cookstoves issued were damaged or have technical issues	Investigate and refer to project lead for possible replacement	As above	5 -7 days
Complaints raised by project staff	Unfair decision related to pay or disciplinary actions	Gather information and refer to project lead	Project manager	5 – 7 days

2.2.11 Feedback and Grievance Redress Procedure Accessibility

The details on procedure for feedback and grievance have been made publicly available to all stakeholders on the project website of Tasman Environmental Markets Asia Pacific <https://www.tasmanenvironmental.com.au/pngcookstoves/>

Additionally, a GRM Standard Operating Procedure (SOP) has also been prepared by Tasman Environmental Markets Asia Pacific and aims to present a set of guidelines for designing and implementing different mechanisms that can address a variety of grievances arising when installing the improved cookstoves in PNG during the crediting period of the project activity. The guidelines are prompt and effective and are timely to inform the complainants of the actions taken to resolve issues raised.

With reference to the GRM, all complaints and grievances will be registered and monitored by the in-country Project Manager supported by the field assistants. These complaints and grievances will be resolved using traditional and customary ways.

This document will be socialized and publicized to communities throughout the project location in their local language, Pidgin English.

2.2.12 Stakeholder Access to Project Documentation

Tasman Environmental Markets Asia Pacific will inform the beneficiaries and their families that they are participating in a project that distributes the G3 Burn rocket stove (also known as Kuniokoa cookstove) upon the household's receipt of the ICS. Tasman Environmental Markets Asia Pacific will also inform them of the need to track their successes and monitor key data, such as money spent on wood fuel and/or time spent collecting wood fuel, among other indicators. ICS end-users will also be made aware that to ensure the integrity of the project and programs, auditors, and stove champions, visit the families up to 2 times after implementation.

As for project documents, the full project documentation will be uploaded on VERRA website, and the same ones will be publicized on the Tasman Environmental Markets Asia Pacific website during the crediting period of the project as the following linkage:

<https://www.tasmanenvironmental.com.au/pngcookstoves/>

2.2.13 Information to Stakeholders on Assessment Process

Tasman Environmental Markets Asia Pacific through its in-country team will inform end-users that they are participating in a project that distributes the ICS free of charge to improve their health, the family economy, and the environment. To make sure that all the stakeholders are aware of the SD vista project assessment, including the site visits of assessors, Tasman Environmental Markets Asia Pacific and its in-country team will inform them in advance (at least 7 days) through the website linkage, phone, email and the following communication channels that are effective in the community.

1. Community announcements through local newspapers and radio *toksaves*

Local newspapers: 'The National and Post courier', and radio shows like *Radio nek blong eagle* (Radio Western Highlands Province) have been used to inform about stakeholder meetings in the past and will be continued. These are paid advertisements those run as community announcements prior to meetings.

2. Facebook Forums and WhatsApp groups

There are several Facebook sites and WhatsApp groups created to discuss developmental issues in each province. For instance, SHP Development forum, EHP Development forum, WHP development forum, among others. This has been utilised for previous stakeholder engagement program details including venues for community engagement, time and date and so forth using both English and tok pidgin (national language).

3. Community meeting platforms and word of mouth

Church gatherings are significant places to inform stakeholders about the project. Community announcements will be prepared in Tok Pidgin and issued to Parish Priests, Pastors and Church elders who make the announcements during their church gatherings. As a significant population are illiterate, phone communications to village elders, ward councillors and project field assistants to spread the information are also an effective method to disseminate information.

The meetings of stakeholder with assessors will be facilitated at a venue informed in advance through the above channels and in the presence of an adequate interpreter (if required due to language restrictions) who would be a community member but not associated to the project.

2.3 Project Management

2.3.1 Avoidance of Corruption

Tasman Environmental Markets Asia Pacific staff worldwide follows strict and robust rules and regulations regarding the avoidance of corruption, code of conduct, and business ethics.

Any person or organization which violates these anti-corruption rules and regulations will immediately stop their work engagement with Tasman Environmental Markets.

2.3.2 Statutory and Customary Rights

The project activity involves distribution of ICSs to individual households only and it will not involve any land use or acquisition.

2.3.3 Recognition of Property Rights

All ICSs covered in this project activity are distributed to individual households free of charge. The end-users will sign a carbon transfer agreement (i.e., End-User Agreement) with Tasman Environmental Markets Asia Pacific to clearly state that the ICS will belong to them, whereas the carbon credits generated from their use will belong to Tasman Environmental Markets.

2.3.4 Free, Prior and Informed Consent

The project is voluntarily implemented by Tasman Environmental Markets, and end-users are free to choose whether they will take be involved in the project or not. Free, prior, and informed consent takes place before distribution through signing of that they have received the ICS, and they will transfer the rights of the carbon credits generated from the project to Tasman Environmental Markets.

2.3.5 Restitution and/or Compensation for Affected Resources

The project activity involves distribution of ICSs to individual households only and it will not affect any resources.

2.3.6 Property Rights Removal/Relocation of Property Rights Holders

The project activity involves distribution of ICSs to individual households only and it will not lead to any removal of property rights or relocation of property rights holders.

2.3.7 Identification of Illegal Activities

Theft and corruption may be identified during the implementation of the project. Tasman Environmental Markets Asia Pacific and its local partner will avoid any corruption as per Section 2.3.1 of this document. End-users will be reminded to keep their ICSs in safe places to reduce theft when they receive the ICSs.

2.3.8 Ongoing Conflicts or Disputes

The project activity involves distribution of ICSs to individual households only. There is no ongoing or unresolved conflicts or disputes over rights to lands, territories and resources and any disputes that were resolved during the last twenty years.

2.3.9 National and Local Laws and Regulations

The project aligns with several International and Papua New Guinean's laws, regulations and conventions Tasman Environmental Markets Asia Pacific, as the project developer, and its in-country team will follow and obey all these relevant laws, regulations and conventions (as stated below).

International Laws, Regulations and Conventions

- **United Nations Framework Convention on Climate Change (UNFCCC,1992)**

The aim of the UNFCCC is to achieve the "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system".

The project has been implemented to reduce Green House Gas (GHG) emissions in an effort to combat climate change, and therefore the Project is compliant with the UNFCCC.

- **United Nations Declaration on the Rights of Indigenous Peoples Rights (2007)**

The Declaration delineates the individual and collective rights of indigenous peoples, their land, territories, resources, culture, identity, language, employment, health, education, and other issues. It also "emphasises the rights of indigenous peoples to maintain and strengthen their own institutions, cultures and traditions, and to pursue their development in keeping with their own needs and aspirations". It "promotes their full and effective participation in all matters that concern them and their right to remain distinct and to pursue their own visions of economic and social development".

The project is aligned with this United Nations Declaration by providing the local (indigenous) communities with the opportunity to engage, at their own free will, in economic development via carbon finance, protection of their lands, territories and forest resources while also enhancing opportunities for other livelihoods, education, employment and healthcare. Furthermore, the project aims to improve the general way of life for current and future

generations, including education on traditional ways of *life* and enhancing community by way of communication, meeting places and knowledge sharing.

- **Stockholm Declaration of the United Nations Conference in Human Environment (1972)**

The Stockholm Declaration states that it is each person's right to freedom, equality, adequate conditions of life; responsibility to protect the environment for present and future generations; to safeguard natural ecosystems and ensure the capacity of the earth to produce renewable resources is maintained, restored, and improved.

The project provides a mechanism whereby the local community and the Papua New Guinean Government can protect the ecosystems and environment of the Project Area, and lead to an improvement in the socio-economic conditions of the community. In addition, the renewable resources of the area will be maintained and possibly improved by the implementation of the project. The socio-economic well-being of local communities will also be improved through income and community development programs.

- **Rio Declaration on Environment and Development (1992)**

The 'Rio' Declaration aims for people to have a healthy and productive life in harmony with nature. It aims to ensure that development activities within national jurisdiction do not cause damage in areas beyond the national boundary. It also aims to eradicate poverty and encourage sustainable development and environmental protection.

The project aims to avoid GHG emissions within Papua New Guinea, which would have had an adverse impact on the global climate extending beyond the country's boundaries. The project also encourages an increase in sustainable livelihoods and education to combat poverty in the project area through sustainable development and environmental protection practices.

- **Paris Agreement (2015)**

The Paris Agreement's goal is to keep the rise in mean global temperature to below 2°C above pre-industrial levels, and preferably limit the increase to 1.5°C to substantially reduce the effects of climate change, and for emissions to be reduced as soon as possible and reach net zero by the middle of the 21st century.

In alignment with this agreement, the project activities reduce GHG by avoiding carbon emissions.

Papua New Guinea's Laws, Regulations and Conventions

- **Papua New Guinea's Constitution (1975)**

The Project is compliant with Papua New Guinea's Constitution, specifically the fourth goal of the Constitution which is "to ensure that the forest resources of the country are used and replenished for the collective benefit of all Papua New Guineans, now and for future generations".

The project aims to reduce the dependence of the people on firewood and in turn avoid deforestation.

- **Climate Change Management Act 2015, Amended (2022), and proposed Carbon Market draft regulation (2023)**

The Act established the Climate Change Development Authority (CCDA) and provided a framework for the development and implementation of measures in Papua New Guinea to combat climate change impacts in accordance with UNFCCC and other international agreements and programs. The project is aligned with the CCDA policies and the proposed carbon market draft regulations.

- **Forestry Act (1991)**

The Act is aimed at managing, developing and protecting Papua New Guinea's forest resources; maximising participation in and development of forest resources as renewable assets; utilising the forest resource to achieve economic growth, employment and industry; and encouraging scientific study and research to contribute to sound ecological balance.

The project assists Papua New Guinea in achieving its goals under the Act to protect and use forest resources through the reduction of firewood consumption and the development of carbon credits; while achieving economic growth through direct employment opportunities, education, and skills development in the project area.

- **Environmental Act (2000)**

The Environmental Act outlines the protection of the environment in accordance with the Fourth National Goal of the Constitution; promotes sustainable development of the environment and economy; states that the social and physical well-being of the people should be supported by the safeguarding the life-supporting capacity of the air, water, soil, and ecosystems for present and future generations, and avoiding, mitigating any adverse effects of activities on the environment.

The project complies with requirements of the Act to promote sustainable development of the environment (protection) and economy (carbon credits, improved cooking) while supporting the social (training, community health) and physical wellbeing (healthcare) of the local communities. The project also reduces the deforestation through reduced demand of firewood.

- **Investment Promotion Act (1992)**

The Papua New Guinea Investment Promotion Authority (IPA) was established by the Investment Promotion Act 1992 to promote and facilitate investment within Papua New Guinea and abroad. The act regulates registration of companies/businesses and the certification of foreign enterprise. The Act states that investors are required to obtain licences for their area of business interest from the relevant Government agencies in the country.

The project is a mechanism for foreign investment in Papua New Guinea's sustainable forests and community development. The project developer has obtained the appropriate licences to operate in their area of businesses.

- **Industrial Relations Act (1962)**

This Act details the settlement of industrial disputes. It is an Act relating to development of industrial relations and the prevention and settlement of industrial disputes.

The project is aligned with this legislation through its own Grievance Redress processes including a formal dispute resolution strategy.

- **Industrial Health Safety and Welfare Act (1962)**

This Act relates to industrial safety, health and welfare, and related purposes. It ensures that measurements have been undertaken to ensure the health, safety, and welfare of employees in the industrial sector.

The project will take all necessary steps to ensure the working health and safety procedures for all the staff involved in the proposed activities.

- **Employment Act No. 54 of 1978 (Consolidated up to 31 March 2001)**

This Act relates to the employment of certain persons and lays out conditions of employment.

The project has employment policies and procedures that comply with the Act.

- **Child Welfare Act (1961)**

The welfare of children and related purposes are covered under this Act, it covers the protection and promotion of the rights and well-being of a child. Project objectives, including activities aimed at improving health standards and improving educational opportunities, are consistent with the Child Welfare Act.

The Project is committed to recognising the rights and welfare of children within the project area.

- **Discriminatory Practices Act (1963)**

This act is to prohibit certain discriminatory practices. 'Discriminatory practice' in this Act means discrimination either of an adverse or of a preferential kind, practiced by a person or group of persons, against or in favour of another person or group of persons for reasons only of colour, race or ethnic, tribal, or national origin.

The project promotes equal opportunities in the workplace based on personal merit rather than gender, colour, race or ethnic, tribal, or national origin.

- **Workers Compensation Act (1978)**

This Act provides for compensation to workers and their dependants in respect of injuries suffered by workers arising out of or in the course of their employment, and for related purposes.

The project is aligned with this act regarding employment policies and procedures related to workers compensation.

- **Papua New Guinea Vision (2050)**

This document maps out the future direction for Papua New Guinea's socio-economic development. It is underpinned by seven 'Strategic Focus Areas', which are referred to as pillars:

- Human Capital Development, Gender, Youth and People Empowerment;
- Wealth Creation;
- Institutional Development and Service Delivery;
- Security and International Relations;
- Environmental Sustainability and Climate Change;
- Spiritual, Cultural and Community Development; and
- Strategic Planning, Integration and Control

In alignment with Papua New Guinea 2050 Vision, the project will develop environmental and sustainability practices and social development and empowerment to the local communities.

2.3.10 Project Ownership

Tasman Environmental Markets Asia Pacific will purchase ICSs and distribute them to end-users free of charge. Before the distribution, end-users and Tasman Environmental Markets Asia Pacific will sign an agreement to confirm that the property right of the ICS belongs to end-users while the carbon credits generated from the project will belong to Tasman Environmental Markets.

2.3.11 Grouped Projects

The Project is a grouped project activity.

New project activities are assessed and evaluated to determine if they provide significant individual impacts (funding of education), household (cookstove), or community level (training on climate change).

For the inclusion of new project activity instances, the project proponent shall ensure that it meets the eligibility criteria below – per Appendix 1 – Grouped Projects (GP) of the SD VISta Standard.

S/N	Criterion	Compliance
1	Adopt and implement the project activities in the same manner as specified in the project description.	New project activity instances will be implemented in the same manner as described in the project description, extending benefits, and reinforcing project stove adoption.
2	Where appropriate, meet the applicability conditions of the SD VISta asset methodology applied to a project.	The project activity, as well as new project instances will apply the SD VISta

		asset methodology and adhere to the criteria set out in the methodology.
3	Are subject to the same scenarios at project start with respect to stakeholders' well-being as determined for initial project instance(s), where (per Section 2.1.5.2 above) the project must meet the criteria of Section 3.1 above.	New project activity instances will have the same baseline scenario, which was available during the start of the project, i.e., usage of G3 Burn rocket stove (also known as Kuniokoa cookstoves) in the households. Also, the instances will be designed to maximize the intended impact and preserve well-being, with monitoring and ongoing stakeholder consultations to ensure impact.
4	Are subject to the same scenarios at project start with respect to natural capital and ecosystem services as determined for initial project instance(s) where (per Section 2.1.5.2 above) projects must meet the criteria of Section 3.2 above.	New project instances will have the same scenario as determined in the baseline scenario. All the new instances will also have households relying on wood fuels for meeting their daily cooking energy demands.
5	Are subject to the same processes for stakeholder engagement described in the project description.	The new project activity instances will be detailed during continuous stakeholder consultations, refresher trainings for Implementing Partners and feedback consider in secondary project activity design to maximize the intended impact and preserve wellbeing.
6	Are subject to the same processes for respect for rights to lands, territories and resources including free, prior and informed consent – described in Section 2.4 above.	The outlined Project Management (Section 2.3) will apply to each additional project activity.
7	Have similar monitoring elements to those set out in the project description.	Additional projects will have equivalent monitoring plans and procedures as described in Section 3.3.

3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1 Condition of Stakeholders at Project Start

(a) The ICS end-users

Social Capital: Health and Well-Being

PNG's annual mean concentration of PM_{2.5} exceeds the recommended maximum of 10 µg/m³, deeming the country's air quality moderately unsafe according to the WHO's guidelines²⁰. Indoor air quality, which affects the health and well-being of men, women, and children, is usually the result of burning of solid fuels for household cooking and heating, and accounts for nearly 2 million deaths annually.

The Baseline Study showed that open fire cooking (i.e. inefficient three-stone fire burning non-renewable woody biomass) produced thick grey or light blue smoke that led to headaches, cough, asthma, and other related diseases.

Social Impact of Firewood Collection:

People living in rural areas across PNG were reported to collect firewood taking an average of 4 trips per week that lasted 3 hours each (ACIAR, 2013). The Baseline Study showed that most of the surveyed population conduct over 3 trips per week for firewood collection.

The use of the ICS in the Highlands region potentially decreased the firewood consumption of up to 1.54 tonnes/household/meal, that in turn reduced the amount of time that families spent collecting firewood. This brings direct economic benefit for those families that purchased less firewood, enables families to have more meaningful for other activities, and contributes to the welfare of the local communities while conserving trees and forests that are vital to tackle the pressing issues of climate change.

Economic Capital: Poverty

Informal sector employment represents around 44.2% of the PNG labour force and is particularly widespread among the youth. A gender gap in access to formal sector employment means that men are more likely to find formal jobs than women. Wages in the informal economy are considerably lower than in the formal economy. This pattern is particularly relevant to the Highlands region that is characterized by low and informal income related to agricultural practices, and 40% of the population is near the poverty line

²⁰ [https://www.who.int/en/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/en/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)

(Hanson, et al., 2001²¹), with an average income estimated as K439/month (Bonilla, et al., 2020). The Highlands region (Hanson, et al., 2001).

Economic Capital: Health and Safety

Informal employment may take the form of small shops operated in residential markets that may include the sale of firewood, which ultimately entails firewood collection. There are significant safety concerns relating to firewood collection by women which may include walking for many hours, facing the danger of assault, and animal attacks. The distribution of improved cookstoves means less firewood is required for cooking and heating, which ultimately reduces the need for regular trips to collect firewood.

Natural Capital: Fuel Baseline

Papua New Guinea's total population of 8.8 million people consumes well over 13 million m³ or 9.4 million tonnes of firewood per year that significantly contributes to Papua New Guinea's Greenhouse Gas (GHG) emissions²². As per the IRENA (2013), over 90% of the population uses firewood for domestic and commercial cooking on traditional open fires. However, the use of firewood geographically related and differs from coastal to high altitude mountain areas. The Highlands region contains 40% of the population in PNG and is characterized by the lowest average temperatures in the country that leads to the highest consumption of firewood for cooking, heating and storytelling in the country (ACIAR, 2013²³; Nuberg, 2015²⁴).

A more detailed analysis of the social and economic capital of the target population in the baseline scenario is mentioned in Section 2.1.8.

(b) Local authorities

Local officers welcome the project because the implementation of project will improve the living standard and economic situation within project boundary.

(c) NGOs

There are various NGOs dedicated to help local people in different ways in Papua New Guinea and are experienced in operating different types of collaborative projects. These NGOs cooperate with Tasman Environmental Markets Asia Pacific to make sure that the project goes well.

3.2 Expected Impacts on Stakeholders

²¹ Hanson, L., Allen, B., Bourke, R. (2001). Papua New Guinea Rural Development Handbook. Canberra, Australian National University

²² <https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2013/Sep/Papua-New-Guinea.pdf?la=en&hash=3E847FD95A91ADAA4CC34614F7A325F80CE36D39>

²³ ACIAR (Australian Centre for International Research). 2013. Promoting diverse firewood production systems in Papua New Guinea

²⁴ Nuberg, I. K. (2015). Developing the fuelwood economy of Papua New Guinea. Energy for Sustainable Development, 24, 9-18.

Impact #1	Provide access to ICS
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	ICS end-users
Resulting Change in Well-being	The project activity aims to replace 100,000 traditional cookstoves with fuel efficient improved cookstoves. Upon gaining access to the ICS, inadequate cooking will be gradually reduced. As will the burden of firewood collection, the time to prepare food and the cost to buy firewood, hence improving health and well-being.

Impact #2	Improved Health Status
Type of Impact	Positive, predicted, direct
Affected Stakeholder Group(s)	ICS end-users
Resulting Change in Well-being	Prior to the project activity, high levels of smoke in the cooking place were observed during the pilot study. Through the use of the ICS, gradually reduce the local people's exposure to high PM2.5 and high CO due to higher efficiency of combustion within their household over the lifetime of the project activity. The monitored baseline emission level is 943.3 mg/MJ, and the project aims to monitor a PM2.5 value lower than the baseline during the project activity through a sampling process during the monitoring period of the project.

Impact #3	Less time spent on unpaid domestic and care work
Type of Impact	Positive, predicted, direct
Affected Stakeholder Group(s)	ICS end-users

Resulting Change in Well-being	Due to higher thermal efficiency, less firewood is required during the cooking process, and time spent cooking is significantly reduced. This results in a direct reduction in time spent on firewood collection and cooking for people. In the baseline, it was found that end-users would collect firewood 3-5 times/week. Upon sampling during the monitoring period, this frequency of collection should reduce as less firewood is required for cooking and heating throughout the week.
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Impact #4	Increased Income
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Local residents, Employees hired for Project Activity
Resulting Change in Well-being	During the project crediting period, Tasman Environmental Markets Asia Pacific will oversee maintenance of cookstoves and the monitoring plan, which will require the need to hire local people. Local people will be hired through a fair and unbiased process, during which we estimate that 130 jobs (10 full-time and 120 part-time) will be created over the project lifespan. The Project targets hiring females as at least 50% of all employees. Employees will be compensated above the average income of K439/month²⁵.

Impact #5	Trainings imparted on climate change, project implementation and monitoring procedures
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	Local community, Beneficiaries, Tasman Environmental Markets Asia Pacific in-country team and Staff

²⁵ Bonilla, F., Voorend, K., Anker, R., Anker, M., & Prates, I. (2020). Anker Living Income Reference Value: Rural Papua New Guinea (2020) (No.20-03-20). Universidad Privada Boliviana.

Resulting Change in Well-being	Prior to the project activity, local residents and end-users had little awareness of ICSs and climate change. With the inception of the project activity, training and skill development related to community engagement, survey conduction, technical trainings like conducting Water Boiling Tests (WBT) will be provided to many stakeholder groups which is envisaged to empower their lives by not only providing long-term employability but also through increased awareness levels regarding issues related to climate change, social equity. This will be done during the onboarding of new staff throughout the lifespan of the project activity.
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Impact #6	Improved Economic Outcomes
Type of Impact	Positive, actual, direct
Affected Stakeholder Group(s)	ICS end-users, ICS end-users' Families
Resulting Change in Well-being	With the commencement of the project activity, the use of the ICS increases fuel efficiency, and an increase in monetary savings is expected. This change is the result of the reduction in the cost of firewood purchases over the lifespan of the project activity due to the increase in fuel efficiency with the project stoves. As per the project activity's Emission Reduction Calculations for project activities in the Highlands, the quantity of firewood calculated to be saved was 3.81tonnes/year. Coupled with the Baseline Survey Report where it was found that end-users typically spent K540/year on firewood, the potential savings an ICS end-user could save is K2,116.8 or USD560.84.

Impact #7	Reduction in sales of Firewood for Firewood Vendors
Type of Impact	Negative, actual, direct
Affected Stakeholder Group(s)	Firewood Vendors

Resulting Change in Well-being

In the absence of the project activity, end-users may opt to procure firewood in order to meet the volumetric need for daily cooking and heating practices.

With the commencement of the project activity, there is a possibility that there will be a reduction in demand of firewood associated with the improved cookstoves that might affect the vendors selling fuel wood in the project area. However, considering the significant demand-supply gap on fuelwood for cooking needs, decrease in the fuel-wood demand in project location is not likely to have a significant negative impact on wood vendors.

3.3 Stakeholder Monitoring Plan

The project activity distributes ICSs to households, which will have a net positive impact on overall well-being of the people in the project area. To account for these positive impacts, a monitoring plan is needed to identify the monitored stakeholder groups, the types of measurements, the sampling methods and the frequency of monitoring and reporting.

Tasman Environmental Market's in-country team will be trained on monitoring techniques that will enable them to collect data through interviews, surveys, direct observations, and group discussions regarding stakeholders' health and employment records. Direct observations and group discussions are design to enable the collection of data that will be compiled into reports that bring integrity and evidence-based credibility to the proposed project activities and will be audited by a third-party specialist.

The monitoring plan for the relevant activities with benefit for People and Prosperity are mentioned in Table 6 below.

Table 6 1: Monitoring plan for activities related to Benefits for People and Prosperity

No.	SDG Target	SDG Indicator	Monitoring Frequency	Parameter Monitored	Evidence
1	1.1	Goal 1: Amount of money saved by families that buy firewood	Annually	Number of cookstoves distributed	Total number of cookstoves distributed. Total number of cookstoves distributed contributes to a reduction of firewood used, which is usually purchased at high prices.
2	3.9	Goal 3: Reduction of fine particulate matter (PM2.5) emissions within households to reduce air pollution within households in the project area.	Annually	Level of fine particulate matter (PM2.5) emissions within households' kitchen area during a cooking activity.	Sampled PM2.5 levels should be below the baseline emission level of 943.3 mg/MJ PM2.5 levels within the kitchen dwelling will be measured using a handheld air quality sensor when the ICS is in use. This reading will be measured for each household selected under the sampling plan during the monitoring period, and the average of all readings will be taken as the final value.
3	4.3	Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Annually or when there are new hires	Number of participants during training sessions	Signed participation/attendance forms, onboarding documents, employment contracts.
	4.4	Goal 4.2: Number of youth and adults with technical and vocational skills, by type of skill within the project area.	Annually	Participation rate is expected to remain constant but may fluctuate slightly depending on the locals available to work.	
4	8.3	Goal 8: Number of people employed, by sector and sex, under the project activity	Annually	Employment contracts	Sampled employment contracts that can be further classified by sector and sex under the project activity.
	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Annually	Employee Payslips	Sampled Employee Payslips should show fair average earning of employees. Data to be categorized by sex, age, occupation, and persons with disabilities.
5	5.4	5.4.1 Proportion of time spent on unpaid	Annually	Number of trips made to collect firewood per week	Sampled trips should be below the baseline scenario of 3-5 times/week.

		domestic and care work, by sex, age, and location.			Reduction of trips made per week indicates more time freed up to do more productive activities such as part-time work, studying, etc.
6	5.5	Goal 5: Number of women in the project including managerial roles	Annually	Number of women hired for the project activity	Employment contracts

Data collected will be stored on a storage cloud.

Initial Visit to Project Area

Prior to the implementation of the project, the in-country team will conduct a one-time identification and awareness visit to local villages and discuss with the leaders of provinces and stakeholders.

They also communicate with other stakeholder such as local officers and NGOs, and search for their support and cooperation. In discussions with stakeholders, they explain the project and its benefits on health, and climate change. Any questions raised are recorded and responded. This will be done in the form of a local stakeholder consultation, that will later feed into the project design documents.

After the project plan is accepted, a community-wide training session is held to introduce the project to all interested households. The training includes information on the multiple benefits and how they can participant in the project.

Vocational and Technical Trainings

Community-wide training sessions will cover key information about the project activity and its lifecycle from distribution to monitoring and everything in-between. To best prepare employees for these tasks, TEM will conduct training workshops prior to distribution that educate employees on the impacts and benefits of the project to equip employees with the knowledge to answer frequently asked questions that may arise during the lifetime of the project activity.

Employees will also be coached on communication procedures to ensure that information relayed to end-users will be transferred according to proper cultural and social understanding, in a clear, articulate and professional manner.

Briefings by technology experts will also be conducted in order to ensure employees are skilled in the installation, maintenance, and repair of the ICS and acquire hands-on practice in assembling and installing improved cookstoves.

These training workshops will be undergone prior to the first distribution visit, then annually as a refresher course for existing employees and new employees being onboarded into the project activity.

Distribution Visit

When distributing the ICSs to end-users, trained in-country staff will explain how to use and maintain the ICSs, respond to any doubts or questions, and recommend the different ways of communication with the local team (e.g., phone, mail, website). Before completing the visit, the in-country team will ensure that the end-users are capable to prepare meals on their new ICSs.

Checking Visits

Tasman Environmental Markets' PNG Team will conduct follow-up visits with households after the distribution. This is the most critical visit because during this time that potential queries regarding the use of the ICS may arise. Villages where the project has been implemented will be chosen randomly. The staff will help end-users facilitate the adaptation, provide support to the women who cook, and solve emerging problems. Feedback from end-users will be collected and analysed with the aim of improving the project activity when applicable, increasing the adaptation of the community to the project.

Verification visit

During each Verification period, Tasman Environmental Markets Asia Pacific and/or the PNG Team will conduct a random sampling survey on the implementation of the project, including all the effects identified in the project's causal chain related to stakeholder well-being. End-users' using experience and cookstove condition will be monitored and recorded in the questionnaires. Other stakeholders' opinion and suggestion will also be recorded. For the measurement of PM 2.5 levels, an Air Quality Monitor will be used to record PM Levels. This will be done on an annual basis during the monitoring period.

3.4 Net Positive Stakeholder Well-being Impacts

The inception and distribution of ICSs undertaken by the project activity will provide a basic service necessary to lead to a healthy and productive life, including saving time and money for wood fuel at the household level. The project will reduce the drudgery undertaken by them especially for women and children through time savings in cutting, collecting, and carrying firewood from trees.

Through the development of the project activity, local stakeholders will yield the following benefits:

1. Reduced Exposure to high CO, PM 2.5 levels and other harmful fumes discharged through inefficient combustion of woody fuel.
2. A higher proportion of population will have access to clean cooking technology for meeting their daily cooking needs.
3. Families will have a greater capacity to save money used for buying wood fuel due to the reduced demand of fuel wood for cooking purposes.
4. Less time will be spent on hard unpaid and dull work of collecting wood fuel and household chores like cooking.

5. Local individuals will be trained and have better technical and vocational skills enhancing their long-term employability and influencing their overall social and economic wellbeing.

4 BENEFITS FOR THE PLANET

4.1 Condition of Natural Capital and Ecosystem Services at Project Start

Papua New Guinea's total population of 8.8 million people consumes well over 13 million m³ or 9.4 million tonnes of firewood per year and in doing so contributes significantly to Papua New Guinea's Greenhouse Gas (GHG) emissions²⁶. As per the IRENA report of 2013²⁷, over 90% of the population occasionally or regularly uses firewood for domestic and commercial cooking on open fires in Papua New Guinea. However, firewood uses vary from area to area or region by region. For example, firewood consumption in the coastal regions may not be same up in the highlands where the altitude is high and temperature is low (Hanson, et al., 2001)²⁸. Firewood use in the highlands is extremely higher than any other regions. As discussed by Nuberg, 2015²⁹, "fuelwood (firewood) is the primary energy source for cooking and heating especially in the highlands where over 40% of the population lives". Therefore, firewood collection and consumed in the highlands is more than the low land areas as the highlanders use firewood not only for cooking but also for heating, lighting, feasting, storytelling and so on.

However, there is little documented understanding of firewood usages, its implications and consumption rate available in the country except a few studies conducted by the Australian Centre for International Agricultural Research (ACIAR) 2013, Murphy (2009) and Page et al. (2016). There is also limited literature to understand how much firewood is used for cooking, heating, and other purposes per household especially across the Highlands region of Papua New Guinea. According to the available literature, Murphy (2009) estimated the average firewood consumption for the Eastern Highlands Province as 3.64 tonnes/household/year, ACIAR (2013) proposed that 7.51 tonnes/household/year was the average firewood consumption nationwide based on data collected from the Highlands region (Eastern Highlands Province, Western Highlands Province and Chimbu Province), National Capital

²⁶ <https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2013/Sep/Papua-New-Guinea.pdf?la=en&hash=3E847FD95A91ADAA4CC34614F7A325F80CE36D39>

²⁷ IRENA (International Renewable Energy Agency), 2013. Renewable energy opportunities and challenges in the Pacific Islands Region <https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2013/Sep/Papua-New-Guinea.pdf?la=en&hash=3E847FD95A91ADAA4CC34614F7A325F80CE36D39>

²⁸ Hanson, L., Allen, B., Bourke, R. (2001). Papua New Guinea Rural Development Handbook. Canberra, Australian National University

²⁹ Nuberg, I. K. (2015). Developing the fuelwood economy of Papua New Guinea. Energy for Sustainable Development, 24, 9-18.

District and Morobe Province, and Page et al. (2016) estimated an average of 11.10 tonnes/household/year for the Morobe Province.

4.2 Expected Impacts on Natural Capital and Ecosystem Services

Impact #1	Reduced demand for large diameter non-renewable firewood from live trees due to only needing finger-sized woody biomass, i.e., twigs and crop residues
Type of Impact	Positive, actual, direct
Affected Natural Capital and/or Ecosystem Service(s)	Live plant biomass, Biodiversity and species Richness, Soil and Water Conservation
Resulting Change in Condition	By slowing the consumption of woody biomass for cooking purposes and by eliminating the need for large diameter woody biomass due to the technology in the ICSs, the resulting drop in demand slows deforestation, thus increasing the above-ground biomass in the adjoining forests near the project area. Reduction in unsustainable harvesting of plant biomass would in turn reduce soil erosion and result in improvement in biodiversity and species richness of the implementation area.

Impact #2	Avoided emission of GHGs made possible by using ICSs
Type of Impact	Positive, predicted, indirect
Affected Natural Capital and/or Ecosystem Service(s)	Air quality in the households as well as surroundings
Resulting Change in Condition	The project stoves require less fuelwood for producing the same amount of thermal energy, therefore, resulting in less carbon dioxide emission in the atmosphere. As carbon dioxide is a greenhouse gas, its reduced emission translates to a positive impact on climate change. An annual average of 242,804tCO₂ is estimated to be avoided due to the inception of the project activity.

4.3 Natural Capital and Ecosystem Services Monitoring Plan

To monitor the impact on natural capital and ecosystem services, Tasman Environmental Markets Asia Pacific in-country team will track:

1. The amount of non-renewable biomass used to meet thermal energy needs for cooking using ICS.
2. Survey the difficulty in collecting firewood, including distance and time.
3. Official data about forest resource, including local, national and international.

The above data will be collected through survey and fuel consumption test, which is the responsibility of Tasman Environmental Markets Asia Pacific in-country team. Tasman Environmental Markets Asia Pacific will train its PNG Team on the monitoring plan and supervise their work. The data may be cross checked by a third-party. The end-user households where the survey and test are conducted will be chosen through random sampling. The data from test and survey will be recorded, analysed and reported by Tasman Environmental Markets. The monitoring and reporting will be conducted each time when verification of SD VISta is conducted.

The monitoring plan for the impacts relevant to benefits for the planet are summarised in Table 7 below.

Table 72: Monitoring plan for activities with benefit for the Planet

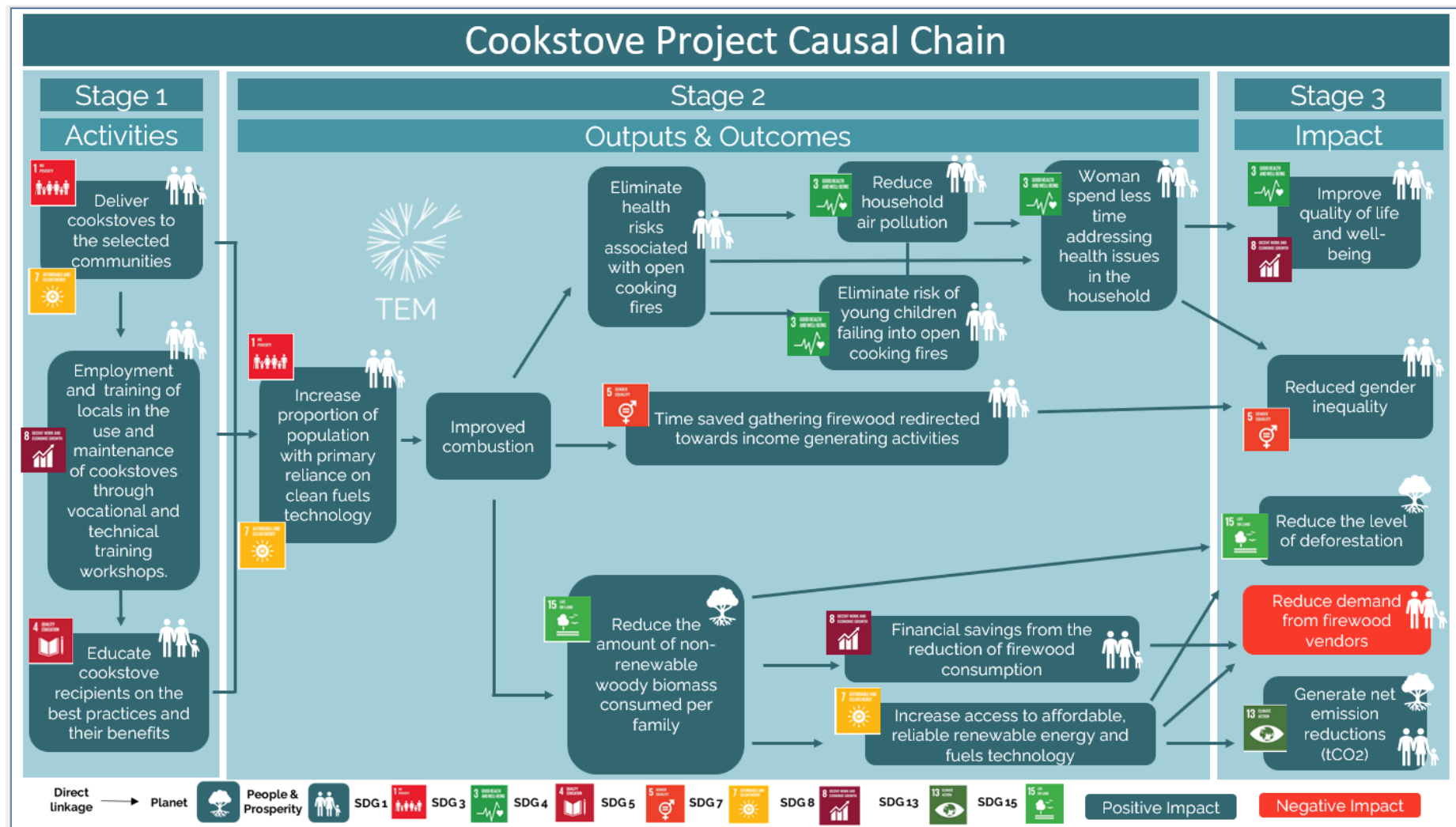
No.	SDG Target	SDG Indicator	Monitoring Frequency	Parameter Monitored	Evidence
1	7.1	Goal 7: Number of people with primary reliance on clean fuels and technology	Annually	Number of cookstoves distributed	Total number of cookstoves distributed will be monitored through purchase invoices that will be validated each time a new batch of ICS is distributed. Total number of cookstoves distributed indicates an increase in a community's access to technology that reduces the consumption of firewood.
2	13.0	Tonnes of greenhouse gas emissions avoided or removed	Annually	Tons of CO2 removed per stove per year	Emission Reduction Calculation Sheet
3	15.2	15.2.1 Progress towards sustainable forest management through the increase of above-ground biomass in forests	Annually	Tons biomass saved per stove per year	Emission Reduction Calculation Sheet

4.4 Net Positive Natural Capital and Ecosystem Services Impacts

By replacing traditional low efficient three-stone fires with high efficiency improved cookstove in households, the project will increase energy efficiency resulting in less firewood combustion, thus generate net GHG reductions.

The average annual GHG emission reduction from the project is expected to be **242,804 tCO_{2e}**. The crediting period is expected to be 10 years. Hence the total GHG emission reduction is **2,428,045 tCO_{2e}**. The project will save **~3.81t** non-renewable biomass every year and it will result in a significant reduction of deforestation.

APPENDIX I: CAUSAL CHAIN



APPENDIX II: LSC OUTREACH

1. Newspaper Advertisements:

There were two newspapers published nationally in PNG. As the project activity would be a grouped project activity, Tasman Environmental Markets Asia Pacific saw the importance of announcing the to the public and informing stakeholders of the project, as well as brief them on the upcoming local stakeholder consultations happening in the Highlands Provinces. The following table shows the date of when these advertisements were published in the newspaper:

S/N	Publishing House	Advertisement Published on: (date)	Supporting Documentation
1	Post Courier Newspaper	14/04/2023	Invoices from respective newspapers for publishing, picture of advertisement in newspapers

2. Letters of Notification to Provincial Governments:

Letters of Notification were also hand delivered to provincial government administrative assistants to brief them on the project activity as well as the consequent co-benefits to their local community that came with the project. Together with the Letter of Notification, a project brief was also attached to aid their understanding. The following table shows the dates of when the letters of no-objection were received from the provincial governments:

S/N	Province	Document Received on: (date)	Supporting Documentation
1	Western Highlands Province	13/02/2023	Letter of No-Objection
2	Jiwaka Province	15/02/2023	
3	Eastern Highlands Province	16/02/2023	
4	Southern Highlands Province	08/03/2023	

During Tasman Environmental Markets's consultations with the provincial governments, the administrative assistants expressed a high level of interest in the project and were supportive of its implementation.

3. Radio Broadcasts

To further Tasman Environmental Market's outreach into the Highlands Provinces regarding the LSCs, radio broadcasts were utilized. All radio broadcasts were done by the National Broadcasting Commission which then broadcasted Tasman Environmental Market's announcements to individual provinces at a frequency of 1 time a day, over 10 days. Each province-level broadcast would then

inform the specific province of the date, location and time of which the LSC was happening within their area. The following table shows the start and end dates of these broadcasts:

S/N	Province	Start of Broadcast: (date)	End of Broadcast: (date)
1	Western Highlands Province	29/02/2023	12/03/2023
2	Jiwaka Province	29/02/2023	12/03/2023

The radio broadcast could only be done for 2 highlands provinces as it was found that the radio stations for the remaining two provinces had been shut down.

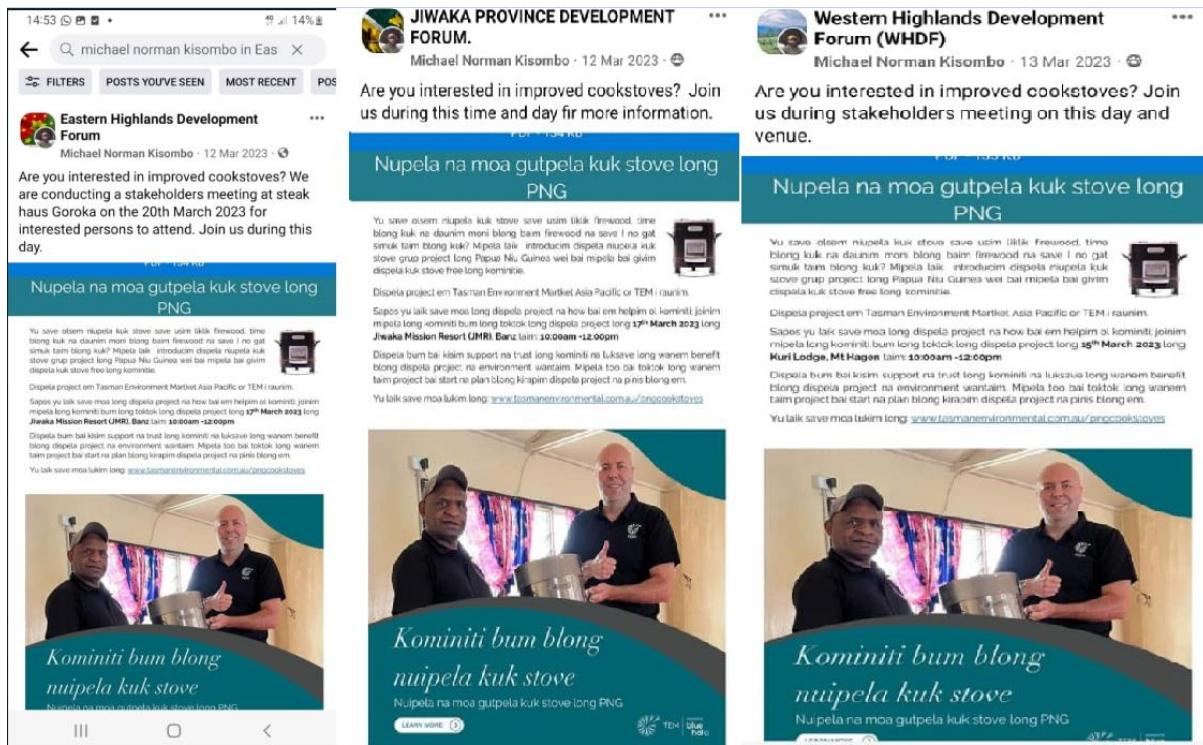
4. Invitation to Churches

Tasman Environmental Markets Asia Pacific is aware that churches and church leaders play an important role in the local community within PNG. As such, invitations to churches were also hand delivered to churches within each province to invite church leaders and elders to participate in the LSC, as well as spread the word about the importance of the LSC within their community by word-of-mouth. The following table shows the dates these invitations were delivered to churches within the provinces:

S/N	Province	Delivery of Invitation: (date)	Supporting Documentation
1	Southern Highlands Province	10/02/2023	Acknowledgement of Receipt by Church Bodies: SHP
2	Western Highlands Province	15/03/2023	Acknowledgement of Receipt by Church Bodies: WHP, Jiwaka, EHP
3	Jiwaka Province	17/03/2023	
4	Eastern Highlands Province	20/03/2023	

5. Social Media Postings

Tasman Environmental Markets Asia Pacific also understood that the majority of the stakeholders received news of events happening within their area through social media. For this, flyers depicting the details of the LSC, and introduction of the project were developed and posted on each province's dedicated Facebook development forums to further boost stakeholder awareness of the LSC. Postings of these flyers started on 12/03/2023.

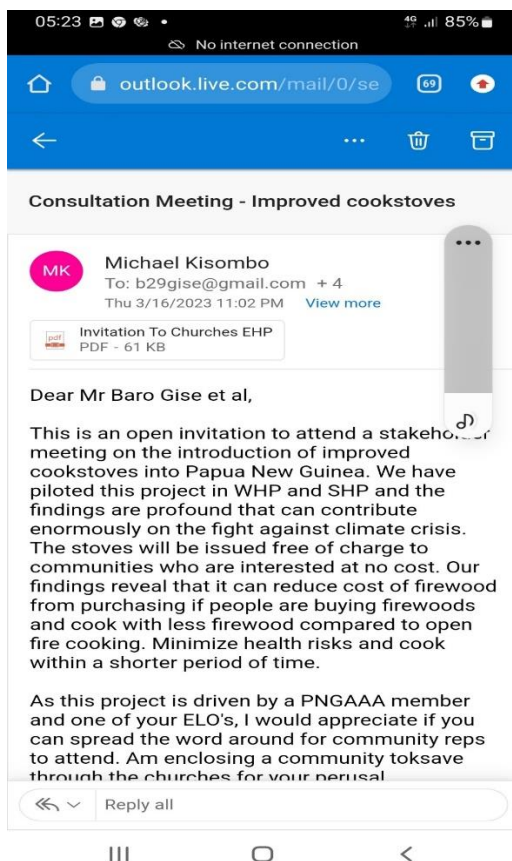
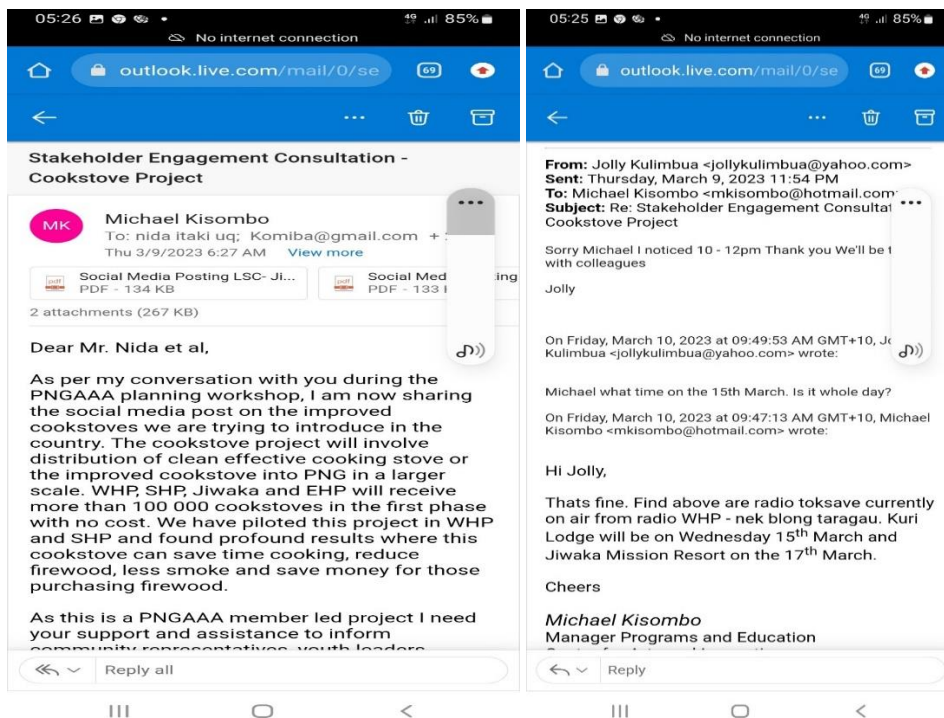


6. Email to PNG Australia Alumni Association (PNGAAA) members

From within Tasman Environmental Market's local partner's connections, Tasman Environmental Markets Asia Pacific also reached out to members from WHP and EHP via email. The table below shows the dates these emails were sent out:

S/N	Province	Date of Email Sent: (date)
1	WHP	09/03/2023
2	EHP	16/03/2023

Screenshots of the emails sent out can also be found below:



APPENDIX III: STAKEHOLDER FEEDBACK

1. Southern Highlands Province:

Questions:

S/N	Question	Answer
1	Ms Julie Ponaria (a teacher from SHP) enquired that there are many big countries contributing to the destruction of the ozone layer and now trying to give (us) cookstoves to stop carbon emissions. She asked if there is a story behind this small cookstove?	Yes, there are many industrial countries that are destroying the ozone layer because they have big factories and industries. We (people of PNG) also do so when cutting down trees for farming and cooking etc... This project is to reduce carbon emissions by using less firewood for daily needs. Hence, saving trees that consume carbon and restore them. That is the story behind this project.
2	Ps Joseph Yapasi of Tente Ward 2 commented that 50 000 cookstoves is not sufficient to cater for all households in Southern Highlands and there may be risks when distribution. Therefore, he suggested that distribution be more targeted. E.g distribute through church networks or distribute to a particular district first and later target other districts.	It was noted for Tasman Environmental Markets Asia Pacific to decide.
3	Ms Regina Kamerepa pointed out that since the cookstoves are going to be issued free, people are likely to rush to get one. Hence, can also steal or pull them from the distributors during the distribution. And asked how Tasman Environmental Markets Asia Pacific would manage that scenario?	This is a threat we are aware of and are working on a strategy on how this can be mitigated during distribution. Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
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1	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
2	Mr. Peter Wari who is in charge of climate change in the province acknowledged the project as it was really going to benefit the community. He further stressed about the importance of the project activity and encouraged the stakeholders to support the project.
3	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/yar) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
4	Another youth mentioned that he provides firewood for three separate houses as most are windows who have no husbands. With the new cookstove, he believes will relieve him from time and struggles he is facing foraging for firewood for all he's mothers.
5	A Health worker emphasized that inhaling of smoke using normal open fire cooking is contributing to a lot of respiratory illnesses and with the introduction of the improved cookstove she believes will minimize such especially mothers and girls.
6	Besides the above reflections and questions, there were quiet a good number of people giving positive comments and supported the project activity. They even asked to increase the volume from 50 000 cookstoves to enough to cater for all households the province.

2. Western Highlands Province:

Questions:

S/N	Question	Answer
1	Mr. Frank Kuri Cr from Mul Baiyer asked, since it's going to affect the rural communities, can the councilors get involved?	Yes, Councilors will get involved to identify heads of households but we do not want this project to be politicized, either be councilors or local MPs.
2	Mr. Nida L Taki, a lecturer at the Agriculture College, Mt Hagen asked if there are further studies made to determine the level of smoke that is given out during normal open fire cooking to see how much carbon emissions are produced.	That's an interesting question. However, this project used observation method to observe the level of smoke when cooking both during the control cooking test and open fire cooking.
3	Mr Dickson Wia stressed that firewood is not only used for cooking in the highlands but for	Yes, we use firewood for many other purposes like for heat, light, storytelling etc. This cookstove is

	other purposes too. Will this cookstove provide for all?	for cooking and it can also heat up the house.
4	Rex Parak enquired if the design of the cookstove can be made locally?	The brand and model of cookstove was chosen specifically for its high efficiency. Tasman Environmental Markets Asia Pacific will consider a local brand in the future if it fits the specifications required to yield the highest efficiency in the combustion of firewood.

Reflections:

S/N	Reflections
1	The District Administrator for Dei District thanked the team for introducing this new innovative idea of cooking as people in his electorate are running out of firewood. He welcomed the project initiative and asked to include all rural communities in his electorate.
2	Those who came from the urban centers also requested that they should be given priority as they are using a lot of money to purchase firewood and its already becoming a burden.
3	One of the participants mentioned that the project is very beneficial as it saves time foraging for firewood and a financial relief in buying firewood.
4	A youth mentioned that this project will benefit the youth as he sees that firewood are already running out.
5	Another supported by saying, the initiative taken by Tasman Environmental Markets Asia Pacific is indeed very vital to reduce carbon monoxide as well as relieving economic burden and reduce health related issues. She suggested that it would be good to involve the councilors.
6	One said, this project must not stop after 10 years. It must continue till the world ends.
7	A health worker supported the project by saying, as a health worker this project if rolled out effectively will help reduce the number of sicknesses like pneumonia, chronic obstructive lung disease and eye diseases as well.

3. Jiwaka Province:

Questions:

S/N	Question	Answer
1	Mr. Lumun Mingal asked how can PNG be compensated well since we are less carbon emitters yet are asked to further reduce the emissions through such cookstove project?	This cookstove project aims to contribute to the wellbeing of the stakeholders in PNG by lessening the burden of purchasing or collecting firewood and improving the respiratory health of end-users exposed to hazardous smoke from the burning of firewood. Cookstoves being distributed for free is paid for through carbon finance and is one method of compensating stakeholders using the carbon credit scheme.

Reflections:

S/N	Reflections
1	A local landowner thanked Tasman Environmental Markets Asia Pacific for this initiative and welcomed everyone to JMR. Having said that he suggested that this project be based there as it is the model village for Jiwaka and must be made the main distribution point.
2	Another said, there are many NGOs like IOM, UNICEF, etc. who come and deliver similar community-based projects but we the people seem to sell those items. He was worried the cookstoves might be sold in a similar manner and appealed to other participants to discourage such practices.
3	A female teacher mentioned that as a teacher she moves from one area to another with kerosine stoves and with the introduction of the improved cooking stove, hers and other teachers would benefit greatly. Not only that but emphasized that they usually ask their students to bring firewood for them once a week and this project will ease that burden.
4	Another person suggested that every household in Jiwaka must receive a cookstove as they are facing difficulties in looking for firewood.
5	Sammy Buka of Banz mentioned that the project is a bottom up where local communities will benefit. He said it's a sickness free project.

6	There were other supportive comments like, Tasman Environmental Markets Asia Pacific must continue this project to other provinces in PNG, delivered through churches, schools and LLG wards.
7	When asked if there are risks involved like those of you selling firewood, would this project affect you? Everyone said no, not at all as they see there is a lot of benefit to the rural communities.
8.	Someone also wanted more than 50 000 cookstoves in Jiwaka as they believe they are good people.

4. Eastern Highlands Province:

Questions:

S/N	Question	Answer
1	Mr Rex Yagusa mentioned that there is a high possibility of the cookstoves being stolen by thieves or destroyed by natural and man-made disasters and asked in such a case, how can we register these complains or cases?	Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	One of the ladies (video graphed) mentioned that this cookstove will lessen the burden for those mothers in the rural setting. She said if they don't sell, there is no money to purchase firewood so they would use plastics and cardboard to cook. This project will really relieve those struggling mothers living in the urban areas.
2	Another suggested having the distribution point centralized and allowing only genuine households to collect. Later field officers can be sent to their locations for registration such as GPS points etc.
3	Someone in the group mentioned that economic benefits are huge as they will be working as the project team to distribute.
4	Those who attended expressed support for the project and demanded if more is distributed in EHP given their population density.

5	No risks or harm mentioned when implementing this project.
6	One participant mentioned that tree-cutting is happening almost every day. More supplies to meet population will help reduce carbon emission.